

	TECHNICAL REPORT NOTA TÉCNICA		No. Núm.	TEA-T-NT-200044
	Department Departamento	Stress Methods & Optimisation	Aircraft Avión	Technologies
Title / Título Program MULTIJOINT User's Manual				
Keywords / Palabras clave SOFTWARE, MANUALS, JOINTS, FASTENERS			Access class Clasificación acceso Airbus Amber P1	
			Revisions Record Registro de Revisiones Pag. 2	
Summary / Resumen This report is the user's manual of program MULTIJOINT, version 1.0, and its purpose is the definition of the way to use the program. MULTIJOINT calculates the distribution of internal loads in a fastened joint between several elements, made of metallic and/or composite materials. External loads may be axial or shear forces at the ends of each element. Thermal stresses are also considered in the calculations.				
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Distribution / Distribución	Execution / Realización		Issue / Edición Issue date / Fecha edición A 8-Jun-2020	
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	Authorisation Autorización			

REGISTRO DE REVISIONES/REVISIONS RECORD

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1 Introduction

This report deals with the user's manual of the program MULTIJOINT, developed by Airbus Defence and Space.

1.1 Technical sheet of the program

Technical Report: TEAS-NT-2010007

PROGRAM

MULTIJOINT

DATE

31-May-2020

OBJECTIVE

MULTIJOINT calculates the distribution of internal loads in a fastened joint between several elements, made of metallic and/or composite materials.

External loads may be axial or shear forces at the ends of each element. Thermal stresses are also considered in the calculations.

LANGUAGE

Fortran 77

Visual Basic .NET

DEVELOPMENT TOOLS

Intel Parallel Studio

MS Visual Studio 2013

OS / PLATFORM

Windows and Linux

Windows GUI

NUMBER OF LINES (APPROX.)

Fortran code: 3680

Visual Basic code: 3940

EXECUTION MODE

Console (input data in text file)

Interactive GUI

AUTHOR

Darío Fernández Villalba

VERSION

1.0

ADDITIONAL DOCUMENTS

See list of applicable references in chapter 1.4.

1.2 Software purpose and description

This report is the user's manual of program MULTIJOINT, version 1.0, and its purpose is the definition of the way to use the program.

MULTIJOINT calculates the distribution of internal loads in different elements of a fastened joint between several elements, made of metallic and/or composite materials.

External loads may be axial or shear forces at the ends of each element. Thermal stresses are also considered in the calculations.

After calculating the internal loads, the program checks the strength of the joined elements and fasteners.

The program performs the following operations:

1. Read a data file prepared by the user, containing the geometric and material properties of the elements to be joined and of the fasteners.
2. Calculate the loads transferred by the fasteners as well as the stress/strain distribution of the elements involved in the joint.
3. Check the strength of the elements involved in the joint.
4. Generate a results file, which includes:
 - The stress/strain distribution in the elements being joined.
 - The loads transferred by the fasteners.
 - Reserve factors of the elements being joined and for the fasteners.

This version is provided with a graphical interface (GUI).

1.3 Abbreviations and symbols

Symbol	Definition
X	X coordinate of geometric points
W	Element Width at X coordinates
t	Element Thickness at X coordinates
E	Material Elasticity modulus
G	Material Shear modulus
μ_1	Material Poisson's ratio in axial direction
μ_2	Material Poisson's ratio in transversal direction
Ct	Thermal expansion coefficient
F_{su}	Ultimate shear stress
F_{tu}	Ultimate tensile stress
F_{bru}	Ultimate bearing stress (only metallic)
A	Allowable by-pass strain around a fastened attachment
B	Allowable bearing stress for the tension-through-the-hole failure
C	Allowable bearing stress for the bearing failure
P_{left}, P_{right}	Axial loads applied at the left/right end of the elements
Q_{left}, Q_{right}	Shear loads applied at the left/right end of the elements
ΔT	Increment of temperature

1.4 References

<i>Reference</i>	<i>Document</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	Stress Methods Manual	1.2	Types of fastened joints. Stress Methods Manual. Chapter 3210.
Ref. 2	Stress Methods Manual	1.0	Bolted Joints in composites. Rivets loads. Stress Methods Manual. Chapter 3610
Ref. 3	Stress Methods Manual	0.1	Strength of fastened joints in composites. Stress Methods Manual. Chapter 3620.

1.5 Differences among versions

Version 1.0

Initial version of the program.

2 Program execution

The program MULTIJOINT can be executed in Windows PCs, in console mode or with GUI (Graphical User Interface), and in Linux server, in console mode.

2.1 Windows versions

The GUI and Console versions are located in the Engineering Directorate server, as well as an electronic copy of this manual and the examples included in chapter 6.

Currently, both executable files are located in the following locations:

- Program with GUI:

<\\Multivac\cae\InfoCalculo\Soft\MultiJoint\MultiJoint.exe>

- Program in console mode:

<\\Multivac\InfoCalculo\Soft\Consola\MultiJoint.exe>

To execute the program, the user has to do double click on the corresponding icon.

In the GUI version, the program shows the following splash (welcome window):



Figure 1: Welcome screen of the Windows GUI version of MULTIJOINT

After this window, the program shows the input data form, as described in chapter 3, to allow to the user intro all the input data or retrieve them from an input file.

For the console version, the execution screen is equal than for the Linux versions and it is the following:

```
MultiJoint 1.0:
=====

Cálculo de la distribución de esfuerzos en una unión
remachada de varios elementos, sometida a cargas casi
uniaxiales en el plano.

Calculation of the stress distribution in a fastened
joint under quasi-uniaxial in-plane loads.

Idioma / Language:
=====

 1 ... ESPAÑOL

 2 ... ENGLISH

Idioma / Language: 2

Name of the input file:   Example.dat

Name of the output file: Example.out
Reading input data file
Calculating stiffness of fasteners
Writing input data
Program in process
Writing output results

MultiJoint 1.0. (c) Airbus Defence and Space. May-2020
```

Figure 2: Interactive execution in console mode

The above run of the program can be executed in on single command, which can be included in a command file (with extension .bat), by passing as arguments the input and output files and the integer to select the language. It would be:

```
\\Multivac\InfoCalculo\Soft\Consola\MultiJoint.exe Example.dat Example.out 2
```

If the third parameter (language) is omitted, the English language will be used.

2.2 Linux version

The Linux Console version of the program is located in the Engineering Directorate server, as well as an electronic copy of this manual and the examples included in chapter 6.

Currently, the executable file is located in the following location:

[/home/locus/cal/Soft_RHEL/multijoint/multijoint.exe](#)

As pointed in previous chapter for the console version, the execution screen is equal for both, Windows and Linux version.

In the same way as for Windows, the above run of the program can be executed in on single command, which can be included in a command file, by passing as arguments the input and output files and the integer to select the language. It would be:

```
/home/locus/cal/Soft_RHEL/multijoint/multijoint.exe Example.dat Example.out 2
```

Again, if the third parameter (language) is omitted, the English language will be used.

Most program calls have been assigned to a command equal than the name of the program. In this case, the above command could be substituted by:

```
multijoint Example.dat Example.out 2
```

3 Input data

The input data can be introduced by means of an input file or providing all data in the corresponding form of the GUI.

3.1 Input file

The data required for program execution can be introduced by means of a text file prepared by the user. The program asks for the complete name of this file and for a filename (without the extension) of the results file. If the data file does not exist, execution ends after emitting the corresponding error message.

3.1.1 Units

The program has been designed to use the following units:

Lengthmillimetres..... mm
ForceNewton N
StressMega Pascal MPa
StrainMicrostrains $\mu\epsilon$
ModulusMega Pascal MPa

If other units are used, problems may appear in the output format of the results file and in the presentation of the input data, even if the program has performed the calculations correctly.

3.1.2 File structure

The data included in the data file is divided into groups. The data items in each line of a data group must be separated by spaces.

The basic structure of all data blocks is the same:

- The first line of each group starts with the group number, which must always start in the first column.

The rest of this line may contain a comment. These comments are not used by the program.

- Following lines included in the group, contain its corresponding data.

In case of error in the input data, the routine reading the input file has been designed to provide an error message, indicating the nature and in which group it is present, to allow later modifications.

The structure of the different groups is described below:

Comments group:

At the beginning of the file, at the end or between two consecutive groups, it is possible insert as many comments group as desired.

Each comments group is a set of lines, all of them starting with an asterisk (*) in the first position of the line.

Group 1 (Title)

The title consists in four lines describing the joint. The data structure of this group is as follows:

Line 1 -----> 1 Optional commentary

Line 2 -----> First line of the title

Line 3 -----> Second line of the title

Line 4 -----> Third line of the title

Line 5 -----> Fourth line of the title

Group 2 (Geometry)

This group defines the X-coordinate of the points where fasteners are located and/or material properties and dimensions of elements are going to be defined (thickness, modulus or width). Its arrangement is:

Line 1 -----> 2 Optional commentary

Line 2 -----> Np

Np Lines --->	1	X(1)
	2	X(2)
	...	
	...	
	Np	X(Np)

Where: X(i) is the X-coordinate of the point "i".

The X-coordinates have to be in ascending order, but a coordinate can be repeated to define a change of thickness.

When a change of any property (e.g. thickness) occurs at a geometric point, two points must be defined with the same X-coordinate, but showing the difference in the property concerned. Figure below shows an example of such duplicated nodes.

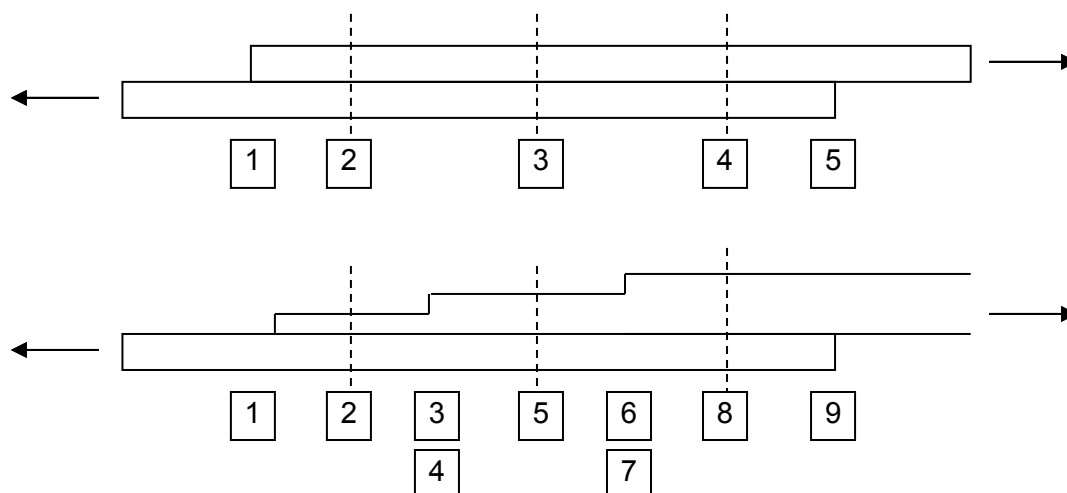


Figure 3: Examples of definitions of point

Group 3 (Fasteners)

This group deals with the fasteners definition. Its data structure is:

Line 1 -----> 3 Optional commentary

Line 2 -----> Ng

Next Lines ->	1	Nr(1)	Ind(1)	Dx(1,1)	D(1,1)	Er(1,1)	Fsu(1,1)
							
							
				Dx(1,Nr)	D(1,Nr)	Er(1,Nr)	Fsu(1,Nr)
	2	Nr(2)	Ind(2)	Dx(2,1)	D(2,1)	Er(2,1)	Fsu(1,1)
							
							
				Dx(2,Nr)	D(2,Nr)	Er(2,Nr)	Fsu(1,Nr)
	...						
	...						
	Ng	Nr(Ng)	Ind(Ng)	Dx(Ng,1)	D(Ng,1)	Er(1,1)	Fsu(1,1)
							
							
				Dx(1,Nr)	D(1,Nr)	Er(1,Nr)	Fsu(1,Nr)

Where: Ng is the number of group of fasteners.

Nr(i) is the number of fasteners included in the group "i".

Ind(i) is the index of the point where the group of fasteners "i" is located. It must be coincident with one of the points defined in group 2.

$Dx(i,j)$ is the Delta of X-coordinate of the fastener "j" inside the group "i".

$$Dx(i,j) = X(\text{Fastener}) - X[(\text{Ind}(i))]$$

$D(i,j)$ is the Diameter of the fastener "j" inside the group "i".

$Er(i,j)$ is the Elasticity modulus of the fastener "j" inside the group "i".

$Fsu(i,j)$ is the ultimate shear stress of the fastener "j" inside the group "i".

Groups of fasteners have to be ordered attending to $\text{Ind}(i)$ parameter, in ascending order.

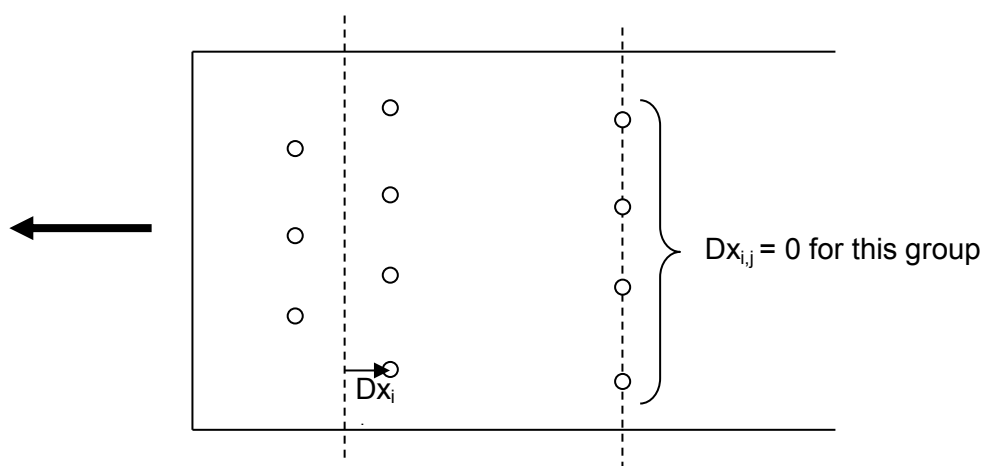


Figure 4: Definition of Dx

Group 4 (Number and type of elements)

This group specifies the number of elements (panels) present in the joint and the material type of each one of them.

Line 1 -----> 4 Optional commentary

Line 2 -----> Nel

Line 3 -----> MatType(1) MatType(2) MatType(Nel)

Where: Nel is the number of elements in the joint.

MatType(i) is the type of material of element "i". It can be:

MatType	Type of material
1	Metallic
2	Composite

Group 5 (Element)

The input file have to include as much groups 5 as elements are present in the joint, one group 5 per element. The structure of this group depends on the type of material of the corresponding element.

Metallic element:

Line 1 -----> 5 Optional commentary

Np Lines ---> 1 W(1) t(1) E(1) G(1) $\mu_1(1)$ $\mu_2(1)$ Ct(1) Ft(1) [Eff(1) Fbru(1)]
 2 W(2) t(2) E(2) G(2) $\mu_1(2)$ $\mu_2(2)$ Ct(2) Ft(2) [Eff(2) Fbru(2)]

 n W(n) t(n) E(n) G(n) $\mu_1(n)$ $\mu_2(n)$ Ct(n) Ft(n) [Eff(n) Fbru(n)]

Composite element:

Line 1 -----> 5 Optional commentary

Np Lines ---> 1 W(1) t(1) E(1) G(1) $\mu_1(1)$ $\mu_2(1)$ Ct(1) $\varepsilon_{\max}(1)$ [Eff(1) A(1) B(1) C(1)]
 2 W(2) t(2) E(2) G(2) $\mu_1(2)$ $\mu_2(2)$ Ct(2) $\varepsilon_{\max}(2)$ [Eff(2) A(2) B(2) C(2)]

 n W(n) t(n) E(n) G(n) $\mu_1(n)$ $\mu_2(n)$ Ct(n) $\varepsilon_{\max}(n)$ [Eff(n) A(n) B(n) C(n)]

Where W(i) is the width of the element at point "i".

t(i) is the thickness of the element at point "i". It can be defined t(i)=0 to create a discontinuity (a cut) in the panel (see chapter 5.3).

E(i) and G(i) are the Elasticity and the Shear moduli of the element at point "i".

$\mu_1(i)$ and $\mu_2(i)$ are the Poisson's ratios, axial and lateral, of the element at point "i".

Ct(i) is the coefficient of thermal expansion of the element at point "i".

Ft(i) is the ultimate tensile stress of the metallic element at point "i".

$\varepsilon_{\max}(i)$ is the maximum allowable strain ($\mu\epsilon$) of the composite element at point "i".

Eff(i) is the effectiveness of the element to transfer load through a fastener at point "i". It must be a number such as:

$$0.0 \leq \text{Eff}(i) \leq 1.0$$

When Eff(i) = 0.0, it means that the fastener is not installed in this element, but in the rest of panels (see chapter 5.2).

Fbru (i) is the ultimate bearing stress of the metallic element at point "i".

A(i), B(i) and C(i) are the allowable values of the element at point "i" for the fastened attachment, related to by-pass stress, bearing for tension-through the hole failure and bearing failure, respectively.

Properties between brackets [] must be included only for locations where a fastener is installed.

Group 6 (Applied loads)

This group defines the applied loads (axial and shear) at both ends of each element as well as the increment of temperature respect to the assembly temperature. Its structure is as follows:

Line 1 -----> 6 Optional commentary

Nel Lines --->	Pleft(1)	Pright(1)	Qleft(1)	Qright(1)
				
				
	Pleft(Nel)	Pright(Nel)	Qleft(Nel)	Qright(Nel)

Line Nel+2 -> ΔT

Where: Pleft(i) and Pright(i) are the axial loads applied at the left and right ends of the element "i".

Qleft(i) and Qright(i) are the shear loads applied at the left and right ends of the element "i".

ΔT is increment of temperature respect to the assembly temperature.

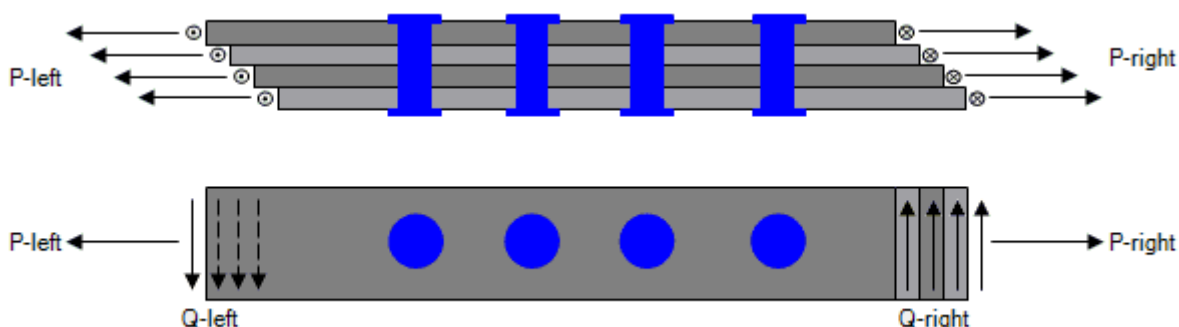


Figure 5: Applied loads

3.2 Windows GUI

The input data can be entered via a file or using the graphical interface. If an input file is used, it is necessary to press the button with the open-icon in the input data form of the program and to choose the name of the file.

Figure 6: “Clean” input data form for three elements (“Geometry” tab)

In the upper-right frame, the user can specify the total number of elements.

The first tab (Geometry) and the first data to introduce in the graphical interface are the number of points required to model the join, and the locations of these points along the X axis.

The second tab (Fasteners) requests for the fasteners of the joint: number of groups of fasteners, locations of the groups and individual fasteners, their diameter, elasticity modulus, shear stiffness (optional), and shear strength.

Next tabs request for the data of the joined elements. They can be metallic or composite laminates. At each point/section, user shall have to introduce thickness, elasticity moduli, coefficient of thermal expansion and allowable strengths (details in following chapter).

The tab before the last one (Sketch) shows a graphical result of the data previously introduced.

Finally, the last tab (Loads) requests for the axial and shear loads applied on the elements ends as well as the increment of temperature.

Figures on next pages show the different tabs of the form.

MultiJoint

Title: EXAMPLE FOR MULTIPLE SHEAR JOINT.
 Fastened double shear joint.
 Adherent 1: Ti. Adherent 2: CFC. Adherent 3: Ti.
 Thermal load: 70°C. Applied load 153000 N.

Elements:
 Number of elements 3

MultiJoint

Geometry Fasteners Element 1 Element 2 Element 3 Sketch Loads

Number of points 16

X

1	0.00
2	30.00
3	60.00
4	60.00
5	75.00
6	90.00
7	90.00
8	105.00
9	120.00
10	120.00
11	135.00
12	150.00
13	150.00
14	180.00

Input data: **Language:**
☐ Español ☒ English

Program execution

Figure 7: Geometry tab: Number of points and their X-coordinates

MultiJoint

Title: EXAMPLE FOR MULTIPLE SHEAR JOINT.
 Fastened double shear joint.
 Adherent 1: Ti. Adherent 2: CFC. Adherent 3: Ti.
 Thermal load: 70°C. Applied load 153000 N.

Elements:
 Number of elements 3

MultiJoint

Geometry Fasteners Element 1 Element 2 Element 3 Sketch Loads

Number of groups of fasteners 3

No. Fast	Point	Delta X	Diameter	Elast Mod	Max Shear		
1	2	5	1	0.00	7.90	115000.00	33112.00
			2	0.00	7.90	115000.00	33112.00
2	2	8	1	0.00	7.90	115000.00	33112.00
			2	0.00	7.90	115000.00	33112.00
3	2	11	1	0.00	7.90	115000.00	33112.00
			2	0.00	7.90	115000.00	33112.00

Input data: **Language:**
☐ Español ☒ English

Program execution

Figure 8: Fasteners tab: Number, locations and properties

MultiJoint

Title: EXAMPLE FOR MULTIPLE SHEAR JOINT.
Fastened double shear joint.
Adherent 1: Ti. Adherent 2: CFC. Adherent 3: Ti.
Thermal load: 70°C. Applied load 153000 N.

Elements:
Number of elements 3

MultiJoint

Geometry Fasteners **Element 1** Element 2 Element 3 Sketch Loads

Material type Metallic

	W	t	E	G	μ _{xy}	μ _{yx}	Alfa	Max. Str.	Eff	Fbru
1	48.00	0.25	115000.0	44230.0	0.30	0.30	8.0000E-06	930.00		
2	48.00	1.50	115000.0	44230.0	0.30	0.30	8.0000E-06	930.00		
3	48.00	1.50	115000.0	44230.0	0.30	0.30	8.0000E-06	930.00		
4	48.00	2.00	115000.0	44230.0	0.30	0.30	8.0000E-06	930.00		
5	48.00	2.00	115000.0	44230.0	0.30	0.30	8.0000E-06	930.00	1.00	1420.00
6	48.00	2.00	115000.0	44230.0	0.30	0.30	8.0000E-06	930.00		
7	48.00	2.50	115000.0	44230.0	0.30	0.30	8.0000E-06	930.00		
8	48.00	2.50	115000.0	44230.0	0.30	0.30	8.0000E-06	930.00	1.00	1420.00
9	48.00	2.50	115000.0	44230.0	0.30	0.30	8.0000E-06	930.00		
10	48.00	3.00	115000.0	44230.0	0.30	0.30	8.0000E-06	930.00		
11	48.00	3.00	115000.0	44230.0	0.30	0.30	8.0000E-06	930.00	1.00	1420.00
12	48.00	3.00	115000.0	44230.0	0.30	0.30	8.0000E-06	930.00		
13	48.00	3.50	115000.0	44230.0	0.30	0.30	8.0000E-06	930.00		
14	48.00	3.50	115000.0	44230.0	0.30	0.30	8.0000E-06	930.00		

Input data: [Icons]
Language: ☐ Español ☒ English [Flags]
Program execution

Figure 9: Metallic element tab: Locations and properties

MultiJoint

Title: EXAMPLE FOR MULTIPLE SHEAR JOINT.
Fastened double shear joint.
Adherent 1: Ti. Adherent 2: CFC. Adherent 3: Ti.
Thermal load: 70°C. Applied load 153000 N.

Elements:
Number of elements 3

MultiJoint

Geometry Fasteners **Element 1** Element 2 Element 3 Sketch Loads

Material type Composite

	W	t	E	G	μ _{xy}	μ _{yx}	Alfa	Max. Def.	Eff	A	B	C
1	48.00	5.40	80518.0	11009.0	0.318	0.119	-4.6300E-0	3650.00				
2	48.00	5.40	80518.0	11009.0	0.318	0.119	-4.6300E-0	3650.00				
3	48.00	5.40	80518.0	11009.0	0.318	0.119	-4.6300E-0	3650.00				
4	48.00	5.40	84466.0	10278.0	0.317	0.106	-5.8600E-0	3650.00				
5	48.00	5.40	84466.0	10278.0	0.317	0.106	-5.8600E-0	3650.00	1.0	4305.0	1120.00	376.00
6	48.00	5.40	84466.0	10278.0	0.317	0.106	-5.8600E-0	3650.00				
7	48.00	5.40	82378.0	11531.0	0.379	0.122	-7.5900E-0	3650.00				
8	48.00	5.40	82378.0	11531.0	0.379	0.122	-7.5900E-0	3650.00	1.0	4364.0	1111.00	376.00
9	48.00	5.40	82378.0	11531.0	0.379	0.122	-7.5900E-0	3650.00				
10	48.00	5.40	91440.0	7616.0	0.228	0.071	-4.7200E-0	3650.00				
11	48.00	5.40	91440.0	7616.0	0.228	0.071	-4.7200E-0	3650.00	1.0	4069.0	1155.00	376.00
12	48.00	5.40	91440.0	7616.0	0.228	0.071	-4.7200E-0	3650.00				
13	48.00	5.40	90389.0	9182.0	0.316	0.088	-7.5100E-0	3650.00				
14	48.00	5.40	90389.0	9182.0	0.316	0.088	-7.5100E-0	3650.00				

Input data: [Icons]
Language: ☐ Español ☒ English [Flags]
Program execution

Figure 10: Composite element tab: Locations and properties

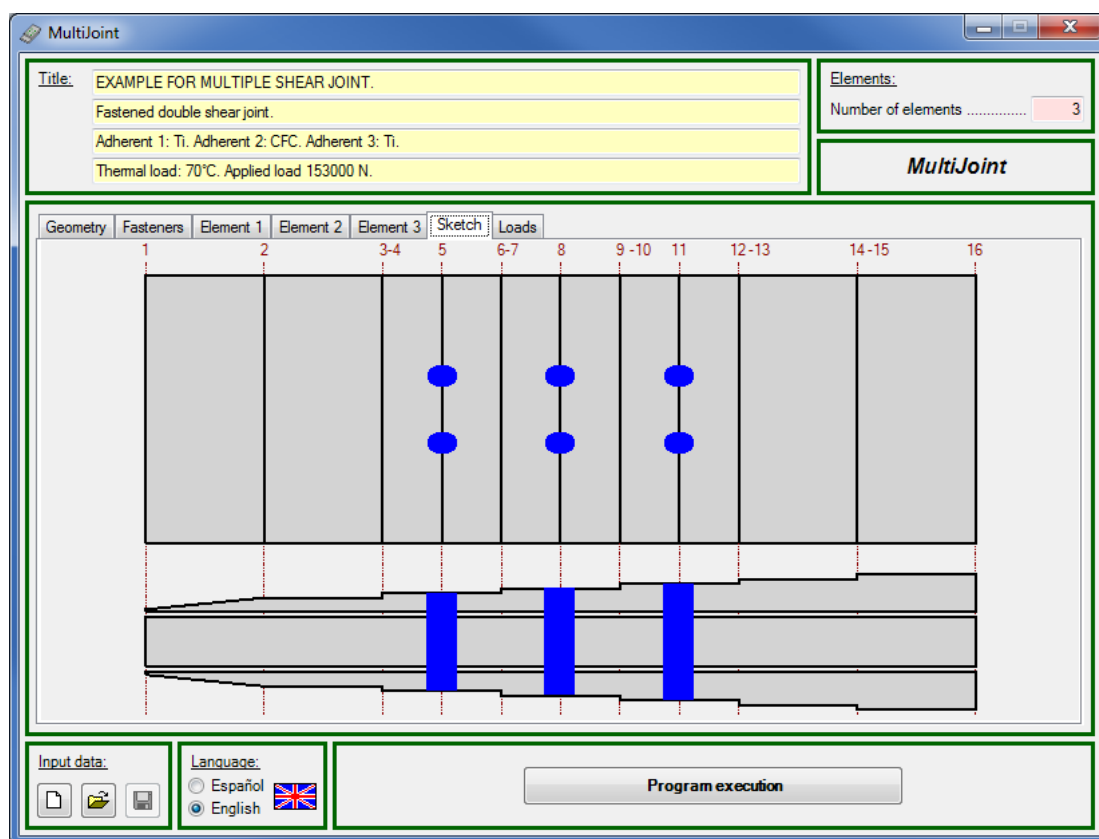


Figure 11: Sketch tab: Graphical interpretation of input data

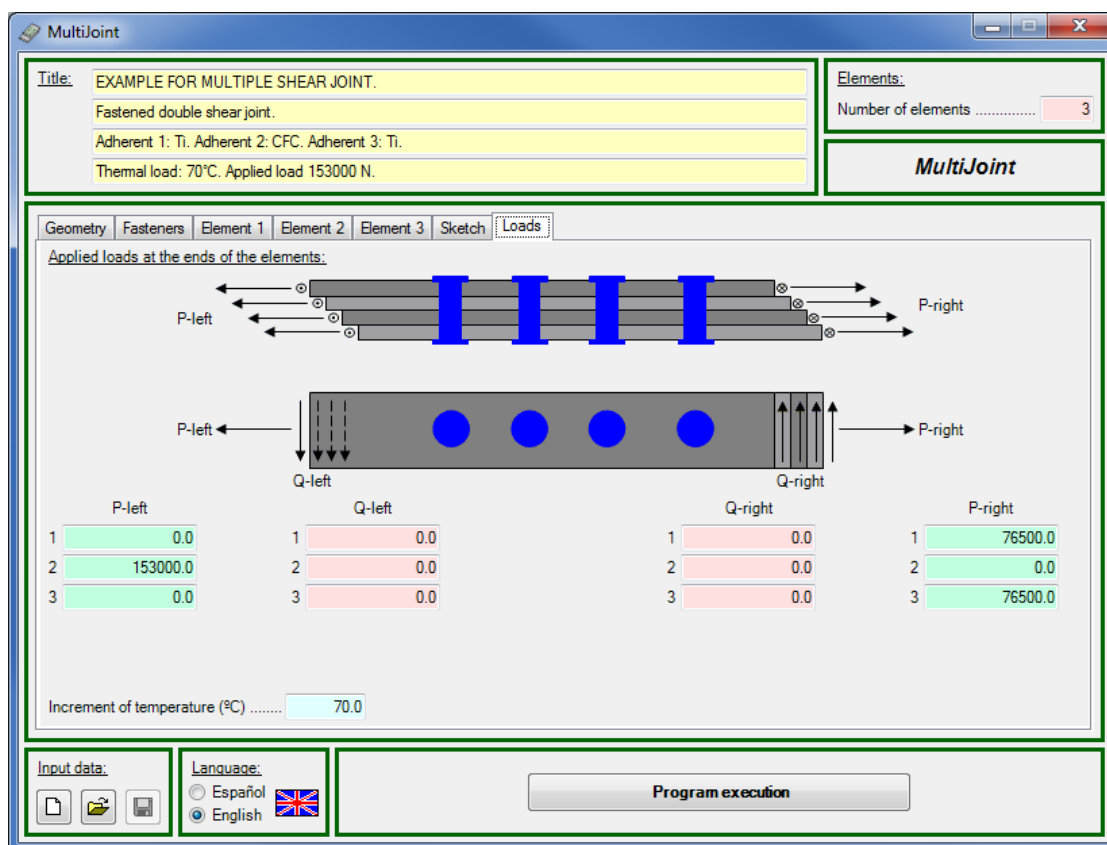


Figure 12: Loads tab: Applied loads

4 Output results

The output results can be obtained in an output file written by the program and/or shown in the graphical user interface GUI.

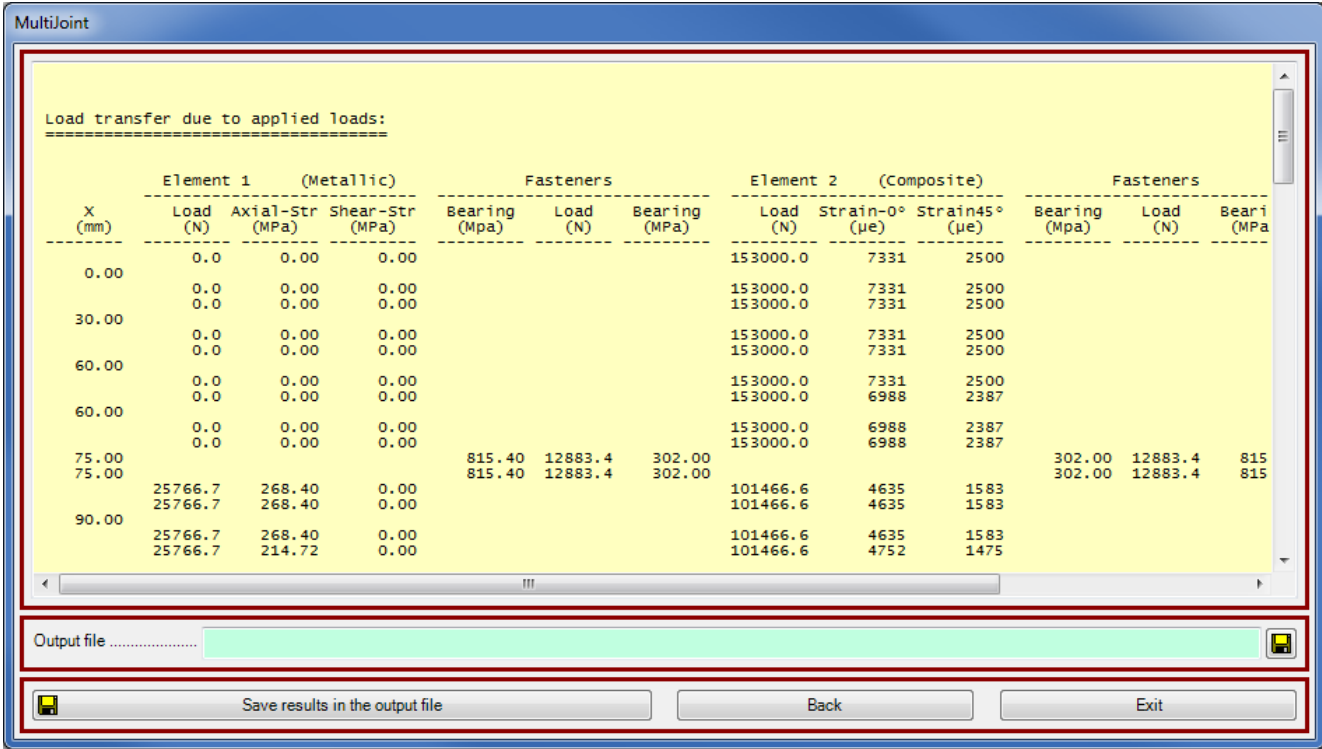
4.1 Output file

This file will contain the following blocks of data:

- An echo of the input data, including all data provided to the program and some previous calculations, in this order:
 - Execution data.
 - Applied loads.
 - Elements data:
 - = Material properties and geometry.
 - = Allowable tensile and bearing values.
 - Fasteners data:
 - = Fasteners locations and properties.
 - = Groups of fasteners global data.
- Load transfer analysis results:
 - Load and stress/strain distribution in elements
 - Transferred load through the fasteners between elements.
 - Bearing stress of elements at fastener locations.
- Reserve factors:
 - Reserve factor for elements:
 - = Unfastened areas. Reserve factor in the most loaded point.
 - = Fastened areas. Reserve factor for each fastener location.
 - Reserve factors for fasteners.
 - = Reserve factor of each fastener between two consecutive elements.

4.2 Windows GUI

In the Windows/PC GUI version, when all data is entered and the button 'Program execution' is pressed, the program runs and a form with the results calculated by the program is shown. As option, those output results can be saved in a text file. The content of this file has been described in previous chapter.



Load transfer due to applied loads:

X (mm)	Element 1 (Metallic)			Fasteners			Element 2 (Composite)			Fasteners		
	Load (N)	Axial-Str (MPa)	Shear-Str (MPa)	Bearing (Mpa)	Load (N)	Bearing (MPa)	Load (N)	Strain-0° (µε)	Strain45° (µε)	Bearing (Mpa)	Load (N)	Beari (MPa)
0.00	0.0	0.00	0.00				153000.0	7331	2500			
	0.0	0.00	0.00				153000.0	7331	2500			
30.00	0.0	0.00	0.00				153000.0	7331	2500			
	0.0	0.00	0.00				153000.0	7331	2500			
60.00	0.0	0.00	0.00				153000.0	7331	2500			
	0.0	0.00	0.00				153000.0	7331	2500			
60.00	0.0	0.00	0.00				153000.0	6988	2387			
	0.0	0.00	0.00				153000.0	6988	2387			
75.00				815.40	12883.4	302.00				302.00	12883.4	815
75.00				815.40	12883.4	302.00				302.00	12883.4	815
	25766.7	268.40	0.00				101466.6	4635	1583			
	25766.7	268.40	0.00				101466.6	4635	1583			
90.00	25766.7	268.40	0.00				101466.6	4635	1583			
	25766.7	214.72	0.00				101466.6	4752	1475			

Output file

Save results in the output file Back Exit

Figure 13: Results form for a MULTIJOINT analysis

4.3 Warning and error messages

The program is designed to give different error messages, which are commonly generated while reading the input data, according to the data block where the error was detected. They are the following ones:

- General errors while reading the input file:
 - The input file cannot be opened
 - Erroneous group number in input file
- Errors while reading the title:
 - The Group 1 has not been defined (Title)
 - End of file while reading Group 1 (Title)
- Errors while reading the geometry:
 - The Group 2 has not been defined (Geometry)
 - The Group 2 has been defined more than once (Geometry)
 - At least two geometry points have to be defined
 - Too many geometry points exceeding the program capability
 - End of file while reading Group 2 (Geometry)
 - Erroneous data in Group 2 (Geometry)
- Errors while reading the fasteners data:
 - The Group 3 has not been defined (Fasteners)
 - The Group 3 has been defined more than once (Fasteners)
 - Group 3 (Fasteners) cannot be defined before Group 2 (Geometry)
 - At least one group of fasteners has to be defined
 - Too many groups of fasteners, exceeding the program capability
 - At least one fastener has to be defined in each group
 - Too many fasteners in a group, exceeding the program capability
 - Fasteners group assigned to a non-existent point
 - End of file while reading Group 3
 - Erroneous data in Group 3 (Fasteners)

- Errors while reading the number and type of elements:
 - The Group 4 has not been defined (N° and type of elements)
 - The Group 4 has been defined more than once '(N° and type of elements)
 - Group 4 (N° and type of elements) cannot be defined before Group 2 (Geometry) and Group 3 (Fasteners)
 - At least two elements have to be defined
 - Too many elements, exceeding the program capability
 - End of file while reading Group 4 (N° and type of elements)
 - Erroneous data in Group 4 (N° and type of elements)
- Errors while reading the elements data:
 - Group 5 has to be defined as many times as elements
 - Group 5 (Elements) cannot be defined before Group 3 (Fasteners)
 - End of file while reading Group 5 (Elements)
 - Erroneous data in Group 5 (Elements)
- Errors while reading loads:
 - The Group 6 has not been defined (Loads)
 - The Group 6 has been defined more than once (Loads)
 - Group 6 (Loads) cannot be defined before Group 4 (N° and type of elements)
 - End of file while reading Group 6 (Loads)
 - Erroneous data in Group 6 (Loads)
- Errors due to incoherence of data:
 - The coordinates of the geometrical points as well as the associated points to fasteners must be in ascending order
 - The applied axial forces are not in equilibrium
 - The applied shear forces are not in equilibrium
- Errors due to problems with the output file:
 - The output file cannot be opened
 - It is not possible to write in the output file

- Errors found during the load transfer analysis:
 - The equations system to determine the load transfer has no solution

5 Software features

5.1 Limitations

To be feasible the analysis of the joint, the following assumptions are made:

- The elements being joined have a linear elastic behaviour
- Fasteners have a linear elastic behaviour, with no variations in their stiffness.
- The fastener stiffness is calculated according to Ref. 1, with the following particularities:
 - If at least one element is composite, the stiffness of each fastener between two consecutive elements is calculated assuming *bolted joint*, even if both elements are metallic.
 - If all elements are metallic, the stiffness of each fastener between two consecutive elements is calculated assuming *riveted joint*.

Furthermore, the following limitations have been included in the program to limit the memory usage:

- Maximum number of elements 9
- Maximum number of geometric points..... 50
- Maximum number of fasteners in a group 5

Note that the maximum number of groups of fasteners is the same as the maximum number of geometric points.

5.2 Non-working fasteners

It is possible to assign an effectiveness of one element to transfer load through a fastener. This is a factor that applies on the stiffness of the joint between two consecutive elements.

In general, the stiffness of the joint between two elements due to a fastener is calculated as:

$$\text{Stiffness} = [\text{Nominal stiffness}] \cdot \text{Effectiveness (Panel 1)} \cdot \text{Effectiveness (Panel 2)}$$

In a particular case where the effectiveness of one element at a fastener location is set to zero, it means that the fastener is not installed in it, but in the rest of panels.

This feature can be better understood in the following example:

- Fasteners:

Two fasteners Installed at points 2 and 3.

- Panels:

Effectiveness of fasteners according to next table:

Panel	Effectiveness	
	Point 2	Point 3
1	1.0	1.0
2	1.0	1.0
3	1.0	0.0
4	1.0	1.0
5	0.0	1.0

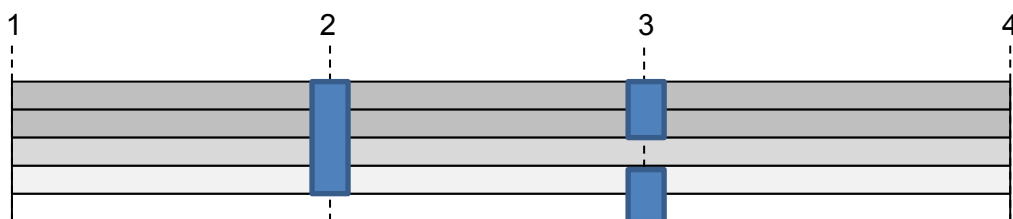


Figure 14: Example of non-working fasteners

5.3 Non-continuous panels

It is possible to split a panel into two or more subpanels, all of them in the same surface as the original one.

To do this, it is necessary to define null thickness ($t = 0$) for the panel at two consecutive points, as shown in next example:

Point	1	2	3	4	5	6	7	8	9	10	11	12
Thickness	1.0	1.0	2.0	2.0	0.0	0.0	2.0	2.0	0.0	0.0	1.0	1.0



Figure 15: Example of non-continuous panels

6 Examples

Two utilisation examples of the program MultiJoint are presented and solved in this chapter. They are also included in the repositories.

6.1 Example1

Double shear load transfer

It consists in a joint between a composite panel, transferring the load in double shear to two metallic panels.

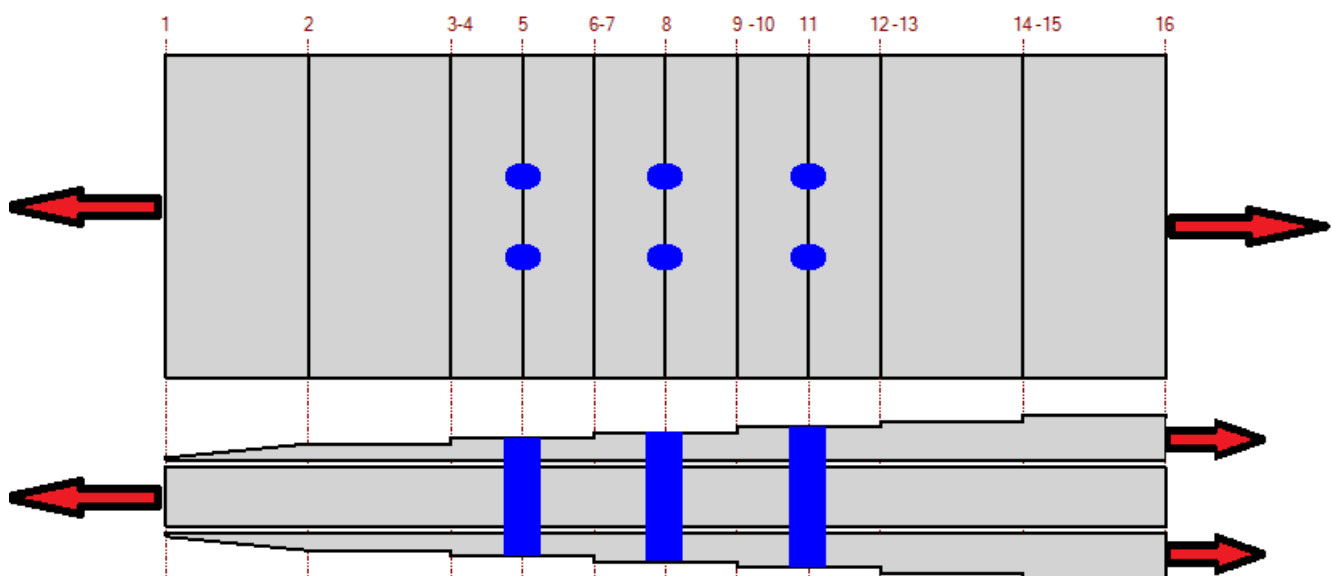


Figure 16:Example 1: Double shear joint

The data items for this case are as follows:

Elements 1 and 3. External panels:

Material: Ti-6Al-4V

Width: $W = 48 \text{ mm}$

Thickness, from left to right: $t = 0.25, 1.5, 2.0, 2.5, 3.0, 3.5$ and 4.0 mm

Elasticity modulus: $E = 115000 \text{ MPa}$

Shear modulus: $G = 44230 \text{ MPa}$

Poisson's ratio: $\mu = 0.3$

Thermal expansion coefficient: $\alpha = 8.0\text{E-}6 \text{ mm}/(\text{mm}\cdot^\circ\text{C})$

Ultimate tensile stress: $F_{tu} = 930 \text{ MPa}$

Ultimate bearing stress: $F_{bru} = 1420 \text{ MPa}$

Element 2. Central panel:

Material: Composite AS4/8552 with variable layup

Width: $W = 48 \text{ mm}$

Thickness: $t = 5.4 \text{ mm}$

Elasticity modulus, from left to right: $E = 80518, 84466, 82378, 91440, 90389 \text{ and } 85477 \text{ MPa}$

Shear modulus, from left to right: $G = 11009, 10278, 11531, 7616, 9182 \text{ and } 12836 \text{ MPa}$

Axial Poisson's ratio, from left to right: $\mu = 0.318, 0.317, 0.379, 0.228, 0.316 \text{ and } 0.639$

Transverse Poisson's ratio, from left to right: $\mu = 0.119, 0.106, 0.122, 0.071, 0.088 \text{ and } 0.127$

Thermal expansion coefficient, from left to right: $\alpha = -4.63\text{E-}7, -5.86\text{E-}7, -7.59\text{E-}7, -4.72\text{E-}7, -7.51\text{E-}7$
and $-1.81\text{E-}6 \text{ mm}/(\text{mm}\cdot^\circ\text{C})$

Ultimate tensile strain: $\varepsilon\text{-max} = 3650 \mu\varepsilon$

Allowable limits for each fastened area: $A = 4305, 4364 \text{ and } 4069 \mu\varepsilon$
 $B = 1120, 1111 \text{ and } 1155 \text{ MPa}$
 $C = 376 \text{ MPa}$

Fasteners:

Pairs of fasteners installed at points 5, 8 and 11 (see Figure 16:).

Diameter: $D=7.9 \text{ mm}$

Elastic modulus: $E_r = 115000 \text{ MPa}$

Allowable shear load: $Sh = 33112 \text{ N}$

Loading:

- Axial loads:

Elements 1 and 3: Loaded each one at their right end with $P_{right} = 76500 \text{ N}$

Element 2: Loaded at its left end with $P_{left} = 153000 \text{ N}$

- Shear loads:

None

- Thermal loads:

An increment of temperature over the assembly temperature of $\Delta T = 70 \text{ }^\circ\text{C}$

The input and output files to solve this example are shown on next pages:

Example 1: Input file

```

1 Title and remarks (4 lines)
  EXAMPLE FOR MULTIPLE SHEAR JOINT.
  Fastened double shear joint.
  Adherent 1: Ti. Adherent 2: CFC. Adherent 3: Ti.
  Thermal load: 70°C. Applied load 153000 N.
2 Geometry: N / i - X(i)
  16
    1      0.00
    2     30.00
    3     60.00
    4     60.00
    5     75.00
    6     90.00
    7     90.00
    8    105.00
    9    120.00
   10    120.00
   11    135.00
   12    150.00
   13    150.00
   14    180.00
   15    180.00
   16    210.00
3 Fasteners data: NG / i  Nr(i)  Ind(i)  Dx(i,k)  D(i,j)  Er(i,j)  Fsu(i,j)
  3
    1  2  5      0.00  7.90  115000.0  33112.00
              0.00  7.90  115000.0  33112.00
    2  2  8      0.00  7.90  115000.0  33112.00
              0.00  7.90  115000.0  33112.00
    3  2 11      0.00  7.90  115000.0  33112.00
              0.00  7.90  115000.0  33112.00
4 Number and type of elements: Nel / Type(i)
  3
  1  2  1
5 Element 1: i  W(i)  t(i)  E(i)  G(i)  µ1(I )  µ2(i)  Ct(i)  Ftu(i)  [Effec Fbru(i)]
  1  48.00  0.25  115000.0  44230.0  0.30  0.30  8.0000E-06  930.00
  2  48.00  1.50  115000.0  44230.0  0.30  0.30  8.0000E-06  930.00
  3  48.00  1.50  115000.0  44230.0  0.30  0.30  8.0000E-06  930.00
  4  48.00  2.00  115000.0  44230.0  0.30  0.30  8.0000E-06  930.00
  5  48.00  2.00  115000.0  44230.0  0.30  0.30  8.0000E-06  930.00  1.000  1420.00
  6  48.00  2.00  115000.0  44230.0  0.30  0.30  8.0000E-06  930.00
  7  48.00  2.50  115000.0  44230.0  0.30  0.30  8.0000E-06  930.00
  8  48.00  2.50  115000.0  44230.0  0.30  0.30  8.0000E-06  930.00  1.000  1420.00
  9  48.00  2.50  115000.0  44230.0  0.30  0.30  8.0000E-06  930.00
 10  48.00  3.00  115000.0  44230.0  0.30  0.30  8.0000E-06  930.00
 11  48.00  3.00  115000.0  44230.0  0.30  0.30  8.0000E-06  930.00  1.000  1420.00
 12  48.00  3.00  115000.0  44230.0  0.30  0.30  8.0000E-06  930.00
 13  48.00  3.50  115000.0  44230.0  0.30  0.30  8.0000E-06  930.00
 14  48.00  3.50  115000.0  44230.0  0.30  0.30  8.0000E-06  930.00
 15  48.00  4.00  115000.0  44230.0  0.30  0.30  8.0000E-06  930.00
 16  48.00  4.00  115000.0  44230.0  0.30  0.30  8.0000E-06  930.00

```


Example 1: Input file (cont.)

```

5 Element 2: i W(i) t(i) E(i) G(i) μ1(i) μ2(i) Ct(i) e-max(i) [Eff A(i) B(i) C(i)]
  1 48.00 5.40 80518.0 11009.0 0.318 0.119 -4.6300E-07 3650.00
  2 48.00 5.40 80518.0 11009.0 0.318 0.119 -4.6300E-07 3650.00
  3 48.00 5.40 80518.0 11009.0 0.318 0.119 -4.6300E-07 3650.00
  4 48.00 5.40 84466.0 10278.0 0.317 0.106 -5.8600E-07 3650.00
  5 48.00 5.40 84466.0 10278.0 0.317 0.106 -5.8600E-07 3650.00 1.000 4305.0
1120.00 376.00
  6 48.00 5.40 84466.0 10278.0 0.317 0.106 -5.8600E-07 3650.00
  7 48.00 5.40 82378.0 11531.0 0.379 0.122 -7.5900E-07 3650.00
  8 48.00 5.40 82378.0 11531.0 0.379 0.122 -7.5900E-07 3650.00 1.000 4364.0
1111.00 376.00
  9 48.00 5.40 82378.0 11531.0 0.379 0.122 -7.5900E-07 3650.00
 10 48.00 5.40 91440.0 7616.0 0.228 0.071 -4.7200E-07 3650.00
 11 48.00 5.40 91440.0 7616.0 0.228 0.071 -4.7200E-07 3650.00 1.000 4069.0
1155.00 376.00
 12 48.00 5.40 91440.0 7616.0 0.228 0.071 -4.7200E-07 3650.00
 13 48.00 5.40 90389.0 9182.0 0.316 0.088 -7.5100E-07 3650.00
 14 48.00 5.40 90389.0 9182.0 0.316 0.088 -7.5100E-07 3650.00
 15 48.00 5.40 85477.0 12836.0 0.639 0.127 -1.8100E-06 3650.00
 16 48.00 5.40 85477.0 12836.0 0.639 0.127 -1.8100E-06 3650.00
5 Element 3: i W(i) t(i) E(i) G(i) μ1(I) μ2(i) Ct(i) Ftu(i) [Eff Fbru(i)]
  1 48.00 0.25 115000.0 44230.0 0.30 0.30 8.0000E-06 930.00
  2 48.00 1.50 115000.0 44230.0 0.30 0.30 8.0000E-06 930.00
  3 48.00 1.50 115000.0 44230.0 0.30 0.30 8.0000E-06 930.00
  4 48.00 2.00 115000.0 44230.0 0.30 0.30 8.0000E-06 930.00
  5 48.00 2.00 115000.0 44230.0 0.30 0.30 8.0000E-06 930.00 1.000 1420.00
  6 48.00 2.00 115000.0 44230.0 0.30 0.30 8.0000E-06 930.00
  7 48.00 2.50 115000.0 44230.0 0.30 0.30 8.0000E-06 930.00
  8 48.00 2.50 115000.0 44230.0 0.30 0.30 8.0000E-06 930.00 1.000 1420.00
  9 48.00 2.50 115000.0 44230.0 0.30 0.30 8.0000E-06 930.00
 10 48.00 3.00 115000.0 44230.0 0.30 0.30 8.0000E-06 930.00
 11 48.00 3.00 115000.0 44230.0 0.30 0.30 8.0000E-06 930.00 1.000 1420.00
 12 48.00 3.00 115000.0 44230.0 0.30 0.30 8.0000E-06 930.00
 13 48.00 3.50 115000.0 44230.0 0.30 0.30 8.0000E-06 930.00
 14 48.00 3.50 115000.0 44230.0 0.30 0.30 8.0000E-06 930.00
 15 48.00 4.00 115000.0 44230.0 0.30 0.30 8.0000E-06 930.00
 16 48.00 4.00 115000.0 44230.0 0.30 0.30 8.0000E-06 930.00
6 Applied loads: Pleft(i) Pright(i) - Qleft(i) Qright(i) / DeltaT
      0.0 76500.0 0.0 0.0
153000.0 0.0 0.0 0.0
      0.0 76500.0 0.0 0.0
      70.0

```

Example 1: Output file

EXAMPLE FOR MULTIPLE SHEAR JOINT.
Fastened double shear joint.
Adherent 1: Ti. Adherent 2: CFC. Adherent 3: Ti.
Thermal load: 70°C. Applied load 153000 N.

Execution data:
=====

Date: 3-Jun-2020
MultiJoint 1.0. (c) Airbus Defence and Space. Jun-2020

Geometry:
=====

Point number	1	2	3	4	·	·	·	Np-3	Np-2	Np-1	Np
Gr. of Fast. number:		1		2	·	·	·	N-1		N	

Element 1: Pleft(1)	<----	#=====	#=====	#=====	#=	·	·	·	#=====	#=====	#=====	#	---->	Pright(1)
Element 2: Pleft(2)	<----	#=====	#=====	#=====	#=	·	·	·	#=====	#=====	#=====	#	---->	Pright(2)
·	·	·	·	·	·	·	·	·	·	·	·	·	·	·
·	·	·	·	·	·	·	·	·	·	·	·	·	·	·
Element n: Pleft(Np)	<----	#=====	#=====	#=====	#=	·	·	·	#=====	#=====	#=====	#	---->	Pright(Np)
			----->	X										

Applied loads (N):
=====

Axial:

Element 1	Pleft =	0.0	Pright =	76500.0	Delta Temp =	70.00 °C
Element 2	Pleft =	153000.0	Pright =	0.0		
Element 3	Pleft =	0.0	Pright =	76500.0		

Shear:

Element 1	Qleft =	0.0	Qright =	0.0
Element 2	Qleft =	0.0	Qright =	0.0
Element 3	Qleft =	0.0	Qright =	0.0

Materials properties and geometry:
=====

```

W    <- Width
t    <- Thickness
E    <- Elastic modulus
G    <- Shear modulus
μxy  <- Poisson ratio <xy>
μyx  <- Poisson ratio <yx>
Alfa <- Thermal expansion coefficient
    
```

Example 1: Output file (cont.)

Element 1 (Metallic)								
Pt	X (mm)	W (mm)	t (mm)	E (MPa)	G (MPa)	μ_{xy}	μ_{yx}	Alfa (mm/mm/C)
1	0.00	48.00	0.25	115000	44230	0.300	0.300	8.000E-06
2	30.00	48.00	1.50	115000	44230	0.300	0.300	8.000E-06
3	60.00	48.00	1.50	115000	44230	0.300	0.300	8.000E-06
4	60.00	48.00	2.00	115000	44230	0.300	0.300	8.000E-06
5	75.00	48.00	2.00	115000	44230	0.300	0.300	8.000E-06
6	90.00	48.00	2.00	115000	44230	0.300	0.300	8.000E-06
7	90.00	48.00	2.50	115000	44230	0.300	0.300	8.000E-06
8	105.00	48.00	2.50	115000	44230	0.300	0.300	8.000E-06
9	120.00	48.00	2.50	115000	44230	0.300	0.300	8.000E-06
10	120.00	48.00	3.00	115000	44230	0.300	0.300	8.000E-06
11	135.00	48.00	3.00	115000	44230	0.300	0.300	8.000E-06
12	150.00	48.00	3.00	115000	44230	0.300	0.300	8.000E-06
13	150.00	48.00	3.50	115000	44230	0.300	0.300	8.000E-06
14	180.00	48.00	3.50	115000	44230	0.300	0.300	8.000E-06
15	180.00	48.00	4.00	115000	44230	0.300	0.300	8.000E-06
16	210.00	48.00	4.00	115000	44230	0.300	0.300	8.000E-06

Element 2 (Composite)								
Pt	X (mm)	W (mm)	t (mm)	E (MPa)	G (MPa)	μ_{xy}	μ_{yx}	Alfa (mm/mm/C)
1	0.00	48.00	5.40	80518	11009	0.318	0.119	-4.630E-07
2	30.00	48.00	5.40	80518	11009	0.318	0.119	-4.630E-07
3	60.00	48.00	5.40	80518	11009	0.318	0.119	-4.630E-07
4	60.00	48.00	5.40	84466	10278	0.317	0.106	-5.860E-07
5	75.00	48.00	5.40	84466	10278	0.317	0.106	-5.860E-07
6	90.00	48.00	5.40	84466	10278	0.317	0.106	-5.860E-07
7	90.00	48.00	5.40	82378	11531	0.379	0.122	-7.590E-07
8	105.00	48.00	5.40	82378	11531	0.379	0.122	-7.590E-07
9	120.00	48.00	5.40	82378	11531	0.379	0.122	-7.590E-07
10	120.00	48.00	5.40	91440	7616	0.228	0.071	-4.720E-07
11	135.00	48.00	5.40	91440	7616	0.228	0.071	-4.720E-07
12	150.00	48.00	5.40	91440	7616	0.228	0.071	-4.720E-07
13	150.00	48.00	5.40	90389	9182	0.316	0.088	-7.510E-07
14	180.00	48.00	5.40	90389	9182	0.316	0.088	-7.510E-07
15	180.00	48.00	5.40	85477	12836	0.639	0.127	-1.810E-06
16	210.00	48.00	5.40	85477	12836	0.639	0.127	-1.810E-06

Example 1: Output file (cont.)

Element 3 (Metallic)								
Pt	X (mm)	W (mm)	t (mm)	E (MPa)	G (MPa)	μ_{xy}	μ_{yx}	Alfa (mm/mm/C)
1	0.00	48.00	0.25	115000	44230	0.300	0.300	8.000E-06
2	30.00	48.00	1.50	115000	44230	0.300	0.300	8.000E-06
3	60.00	48.00	1.50	115000	44230	0.300	0.300	8.000E-06
4	60.00	48.00	2.00	115000	44230	0.300	0.300	8.000E-06
5	75.00	48.00	2.00	115000	44230	0.300	0.300	8.000E-06
6	90.00	48.00	2.00	115000	44230	0.300	0.300	8.000E-06
7	90.00	48.00	2.50	115000	44230	0.300	0.300	8.000E-06
8	105.00	48.00	2.50	115000	44230	0.300	0.300	8.000E-06
9	120.00	48.00	2.50	115000	44230	0.300	0.300	8.000E-06
10	120.00	48.00	3.00	115000	44230	0.300	0.300	8.000E-06
11	135.00	48.00	3.00	115000	44230	0.300	0.300	8.000E-06
12	150.00	48.00	3.00	115000	44230	0.300	0.300	8.000E-06
13	150.00	48.00	3.50	115000	44230	0.300	0.300	8.000E-06
14	180.00	48.00	3.50	115000	44230	0.300	0.300	8.000E-06
15	180.00	48.00	4.00	115000	44230	0.300	0.300	8.000E-06
16	210.00	48.00	4.00	115000	44230	0.300	0.300	8.000E-06

Allowable tensile and bearing values for elements:

=====

Element 1 (Metallic)					
Tensile			Bearing		
Pt	X (mm)	Max Str (MPa)	Gr	Nr	Fbru (MPa)
1	0.00	930.00	1	1	1420.00
2	30.00	930.00		2	1420.00
3	60.00	930.00	2	1	1420.00
4	60.00	930.00		2	1420.00
5	75.00	930.00	3	1	1420.00
6	90.00	930.00		2	1420.00
7	90.00	930.00			
8	105.00	930.00			
9	120.00	930.00			
10	120.00	930.00			
11	135.00	930.00			
12	150.00	930.00			
13	150.00	930.00			
14	180.00	930.00			
15	180.00	930.00			
16	210.00	930.00			

Example 1: Output file (cont.)

Element 2 (Composite)							
Tensile			Bearing				
Pt	X (mm)	Max Def (µe)	Gr	Nr	A (µe)	B (MPa)	C (MPa)
1	0.00	3650	1	1	4305	1120.00	376.00
2	30.00	3650		2	4305	1120.00	376.00
3	60.00	3650	2	1	4364	1111.00	376.00
4	60.00	3650		2	4364	1111.00	376.00
5	75.00	3650	3	1	4069	1155.00	376.00
6	90.00	3650		2	4069	1155.00	376.00
7	90.00	3650					
8	105.00	3650					
9	120.00	3650					
10	120.00	3650					
11	135.00	3650					
12	150.00	3650					
13	150.00	3650					
14	180.00	3650					
15	180.00	3650					
16	210.00	3650					

Element 3 (Metallic)					
Tensile			Bearing		
Pt	X (mm)	Max Str (MPa)	Gr	Nr	Fbru (MPa)
1	0.00	930.00	1	1	1420.00
2	30.00	930.00		2	1420.00
3	60.00	930.00	2	1	1420.00
4	60.00	930.00		2	1420.00
5	75.00	930.00	3	1	1420.00
6	90.00	930.00		2	1420.00
7	90.00	930.00			
8	105.00	930.00			
9	120.00	930.00			
10	120.00	930.00			
11	135.00	930.00			
12	150.00	930.00			
13	150.00	930.00			
14	180.00	930.00			
15	180.00	930.00			
16	210.00	930.00			

Example 1: Output file (cont.)

Fasteners locations: 3
=====

Gr	Pt	Nr	Delta X (mm)	Diam (mm)	Er (MPa)	Allowable (N)	Stiffness (N/mm)	Effectiveness (%)	Elem.
1	5	1	0.00	7.90	115000	33112.00	48384.78	100.0 - 100.0	1 - 2
							48384.77	100.0 - 100.0	2 - 3
		2	0.00	7.90	115000	33112.00	48384.78	100.0 - 100.0	1 - 2
							48384.77	100.0 - 100.0	2 - 3
2	8	1	0.00	7.90	115000	33112.00	53315.12	100.0 - 100.0	1 - 2
							53315.12	100.0 - 100.0	2 - 3
		2	0.00	7.90	115000	33112.00	53315.12	100.0 - 100.0	1 - 2
							53315.12	100.0 - 100.0	2 - 3
3	11	1	0.00	7.90	115000	33112.00	58965.94	100.0 - 100.0	1 - 2
							58965.94	100.0 - 100.0	2 - 3
		2	0.00	7.90	115000	33112.00	58965.94	100.0 - 100.0	1 - 2
							58965.94	100.0 - 100.0	2 - 3

Groups of fasteners. Global data:
=====

Gr	Pt	C.G. (mm)	Stiffness (N/mm)	Elem.
1	5	75.00	96769.56	1 - 2
		75.00	96769.55	2 - 3
2	8	105.00	106630.25	1 - 2
		105.00	106630.25	2 - 3
3	11	135.00	117931.88	1 - 2
		135.00	117931.88	2 - 3

Example 1: Output file (cont.)

Load transfer due to applied loads:														
Element 1 (Metallic)				Fasteners			Element 2 (Composite)			Fasteners			Element 3 (Metallic)	
X (mm)	Load (N)	Axial-Str (MPa)	Shear-Str (MPa)	Bearing (MPa)	Load (N)	Bearing (MPa)	Strain-0° (µε)	Strain-45° (µε)	Load (N)	Bearing (MPa)	Load (N)	Bearing (MPa)	Axial-Str (MPa)	Shear-Str (MPa)
0.00	0.0	0.00	0.00				7331	2500	153000.0				0.0	0.00
30.00	0.0	0.00	0.00				7331	2500	153000.0				0.0	0.00
	0.0	0.00	0.00				7331	2500	153000.0				0.0	0.00
	0.0	0.00	0.00				7331	2500	153000.0				0.0	0.00
60.00	0.0	0.00	0.00				7331	2500	153000.0				0.0	0.00
60.00	0.0	0.00	0.00				6988	2387	153000.0				0.0	0.00
75.00	0.0	0.00	0.00				6988	2387	153000.0				0.0	0.00
75.00	25766.7	268.40	0.00			302.00	4635	1583	101466.6		302.00	12883.4	815.40	0.00
90.00	25766.7	268.40	0.00			302.00	4635	1583	101466.6		302.00	12883.4	815.40	0.00
90.00	25766.7	214.72	0.00				4635	1583	101466.6					0.00
105.00	25766.7	214.72	0.00				4752	1475	101466.6					0.00
105.00	25766.7	214.72	0.00				4752	1475	101466.6					0.00
120.00	47802.6	398.35	0.00				2688	835	57394.9					0.00
120.00	47802.6	398.35	0.00				2688	835	57394.9					0.00
120.00	47802.6	331.96	0.00				2422	935	57394.9					0.00
135.00	47802.6	331.96	0.00				2422	935	57394.9					0.00
135.00	47802.6	331.96	0.00				2422	935	57394.9					0.00
150.00	76500.0	531.25	0.00				0	0	0.0					0.00
150.00	76500.0	531.25	0.00				0	0	0.0					0.00
150.00	76500.0	531.25	0.00				0	0	0.0					0.00
150.00	76500.0	455.36	0.00				0	0	0.0					0.00
180.00	76500.0	455.36	0.00				0	0	0.0					0.00
180.00	76500.0	455.36	0.00				0	0	0.0					0.00
180.00	76500.0	398.44	0.00				0	0	0.0					0.00
210.00	76500.0	398.44	0.00				0	0	0.0					0.00
210.00	76500.0	398.44	0.00				0	0	0.0					0.00

Example 1: Output file (cont.)

Reserve factors for applied loads:
=====

Unfastened areas:

	Element 1	Element 2	Element 3
	-----	-----	-----
Material:	Metallic	Composite	Metallic
Most loaded point:	150.00 mm	60.00 mm	150.00 mm
Von-Mises stress:	531.25 MPa		531.25 MPa
Maximum strain:		7331 µe	
Allowable Stress/strain:	930.00 MPa	3650 µe	930.00 MPa
Reserve factor:	1.75	0.49	1.75

Fastened areas:

Element 1				

Gr	Nr	Bearing (MPa)	Fbru (MPa)	R.F.
---	---	-----	-----	-----
1	1	815.40	1420.00	1.74
	2	815.40	1420.00	1.74
2	1	557.87	1420.00	>2.00
	2	557.87	1420.00	>2.00
3	1	605.43	1420.00	>2.00
	2	605.43	1420.00	>2.00

Element 2							

Gr	Nr	By-pass (µe)	Bearing (MPa)	A (µe)	B (MPa)	C (MPa)	R.F.
---	---	-----	-----	-----	-----	-----	-----
1	1	4635	427.09	4305	1120.00	376.00	0.68
	2	4635	427.09	4305	1120.00	376.00	0.68
2	1	2688	365.25	4364	1111.00	376.00	1.02
	2	2688	365.25	4364	1111.00	376.00	1.02
3	1	0	475.67	4069	1155.00	376.00	0.79
	2	0	475.67	4069	1155.00	376.00	0.79

Element 3				

Gr	Nr	Bearing (MPa)	Fbru (MPa)	R.F.
---	---	-----	-----	-----
1	1	815.40	1420.00	1.74
	2	815.40	1420.00	1.74
2	1	557.87	1420.00	>2.00
	2	557.87	1420.00	>2.00
3	1	605.43	1420.00	>2.00
	2	605.43	1420.00	>2.00

Example 1: Output file (cont.)

Reserve factors for fasteners:
=====

Gr	Nr	Joint	Shear (N)	Allowable (N)	R.F.
---	---	-----	-----	-----	-----
1	1	1-2	12883.4	33112.0	>2.00
		2-3	12883.4	33112.0	>2.00
	2	1-2	12883.4	33112.0	>2.00
		2-3	12883.4	33112.0	>2.00
2	1	1-2	11017.9	33112.0	>2.00
		2-3	11017.9	33112.0	>2.00
	2	1-2	11017.9	33112.0	>2.00
		2-3	11017.9	33112.0	>2.00
3	1	1-2	14348.7	33112.0	>2.00
		2-3	14348.7	33112.0	>2.00
	2	1-2	14348.7	33112.0	>2.00
		2-3	14348.7	33112.0	>2.00

6.2 Example2

Staggered flux repair

It consists in a joint between four metallic panels. Only the upper panel is loaded at both ends and it is split in three parts. This configuration simulates a staggered flux repair, where the upper panel has been cut, removed and substituted by the upper central panel.

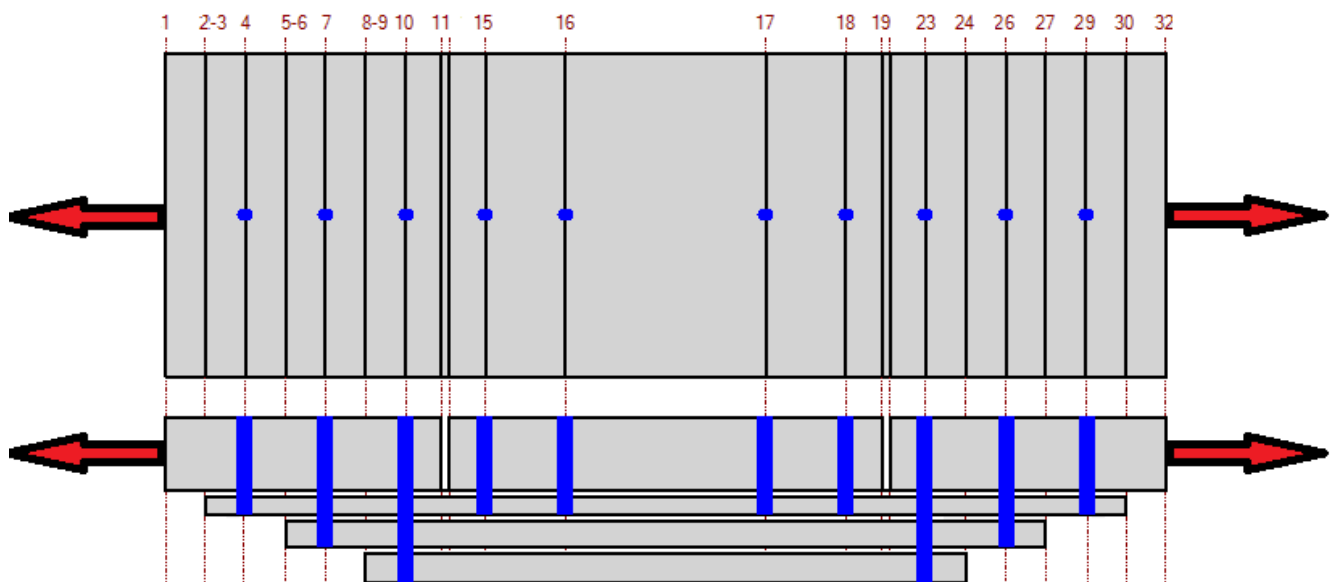


Figure 17:Example 2: Staggered flux repair

The data items for this case are as follows:

Elements 1 to 4:

Material: Aluminium 2024 T3

Width: $W = 50 \text{ mm}$

Thickness
 Element 1: $t = 2.0 \text{ mm}$
 Element 2: $t = 0.5 \text{ mm}$
 Element 3: $t = 0.7 \text{ mm}$
 Element 4: $t = 0.8 \text{ mm}$

Elasticity modulus: $E = 72400 \text{ MPa}$

Shear modulus: $G = 27600 \text{ MPa}$

Poisson's ratio: $\mu = 0.33$

Thermal expansion coefficient: $\alpha = 2.25\text{E-}5 \text{ mm}/(\text{mm}\cdot^\circ\text{C})$

Ultimate tensile stress: $F_{tu} = 450 \text{ MPa}$

Ultimate bearing stress: $F_{bru} = 900 \text{ MPa}$

Fasteners:

Single fasteners installed at points 4, 7, 19, 15, 16, 17, 18, 23, 26 and 29.

Fasteners at locations 15, 16, 17 and 18 joint only elements 1 and 2 (see Figure 17:Figure 16:).

Diameter: $D = 4.0 \text{ mm}$

Elastic modulus: $E_r = 110000 \text{ MPa}$

Allowable shear load: $Sh = 33000 \text{ N}$

Loading:

- Axial loads:

Elements 1: Loaded at both ends with $P_{left} = P_{right} = 10000 \text{ N}$

- Shear loads:

None

- Thermal loads:

No increment of temperature over the assembly temperature: $\Delta T = 0 \text{ }^{\circ}\text{C}$

The input and output files to solve this example are shown on next pages:

Example 2: Input file

```

1 Title and remarks (4 lines)
  Program MultiJoint
  Validation example with four panels fastened between them. Upper one split in
  Material: Aluminium 2024 T3.
  Upper panel loaded at both ends.
2 Geometry: N / i - X(i)
  32
    1      0.00
    2     10.00
    3     10.00
    4     20.00
    5     30.00
    6     30.00
    7     40.00
    8     50.00
    9     50.00
   10     60.00
   11     69.00
   12     69.00
   13     71.00
   14     71.00
   15     80.00
   16    100.00
   17    150.00
   18    170.00
   19    179.00
   20    179.00
   21    181.00
   22    181.00
   23    190.00
   24    200.00
   25    200.00
   26    210.00
   27    220.00
   28    220.00
   29    230.00
   30    240.00
   31    240.00
   32    250.00
3 Fasteners data: NG / i  Nr(i)  Ind(i)  Dx(i,k)  D(i,j)  Er(i,j)  Fsu(i,j)
  10
    1   1   4   0.00   4.00  110000.0  33000.00
    2   1   7   0.00   4.00  110000.0  33000.00
    3   1  10   0.00   4.00  110000.0  33000.00
    4   1  15   0.00   4.00  110000.0  33000.00
    5   1  16   0.00   4.00  110000.0  33000.00
    6   1  17   0.00   4.00  110000.0  33000.00
    7   1  18   0.00   4.00  110000.0  33000.00
    8   1  23   0.00   4.00  110000.0  33000.00
    9   1  26   0.00   4.00  110000.0  33000.00
   10   1  29   0.00   4.00  110000.0  33000.00
4 Number and type of elements: Nel / Type(i)
  4
  1  1  1  1

```

Example 2: Input file (cont.)

5	Element	1:	i	W(i)	t(i)	E(i)	G(i)	$\mu_1(i)$	$\mu_2(i)$	Ct(i)	Ftu(i)	[Eff	Fbru(i)]
	1		50.00	2.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	2		50.00	2.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	3		50.00	2.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	4		50.00	2.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00	1.000	900.00	
	5		50.00	2.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	6		50.00	2.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	7		50.00	2.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00	1.000	900.00	
	8		50.00	2.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	9		50.00	2.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	10		50.00	2.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00	1.000	900.00	
	11		50.00	2.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	12		50.00	0.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	13		50.00	0.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	14		50.00	2.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	15		50.00	2.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00	1.000	900.00	
	16		50.00	2.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00	1.000	900.00	
	17		50.00	2.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00	1.000	900.00	
	18		50.00	2.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00	1.000	900.00	
	19		50.00	2.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	20		50.00	0.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	21		50.00	0.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	22		50.00	2.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	23		50.00	2.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00	1.000	900.00	
	24		50.00	2.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	25		50.00	2.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	26		50.00	2.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00	1.000	900.00	
	27		50.00	2.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	28		50.00	2.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	29		50.00	2.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00	1.000	900.00	
	30		50.00	2.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	31		50.00	2.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	32		50.00	2.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			

Example 2: Input file (cont.)

5	Element	2:	i	W(i)	t(i)	E(i)	G(i)	$\mu_1(i)$	$\mu_2(i)$	Ct(i)	Ftu(i)	[Eff	Fbru(i)]
	1		50.00	0.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	2		50.00	0.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	3		50.00	0.50	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	4		50.00	0.50	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00	1.000	900.00	
	5		50.00	0.50	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	6		50.00	0.50	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	7		50.00	0.50	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00	1.000	900.00	
	8		50.00	0.50	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	9		50.00	0.50	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	10		50.00	0.50	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00	1.000	900.00	
	11		50.00	0.50	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	12		50.00	0.50	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	13		50.00	0.50	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	14		50.00	0.50	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	15		50.00	0.50	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00	1.000	900.00	
	16		50.00	0.50	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00	1.000	900.00	
	17		50.00	0.50	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00	1.000	900.00	
	18		50.00	0.50	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00	1.000	900.00	
	19		50.00	0.50	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	20		50.00	0.50	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	21		50.00	0.50	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	22		50.00	0.50	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	23		50.00	0.50	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00	1.000	900.00	
	24		50.00	0.50	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	25		50.00	0.50	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	26		50.00	0.50	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00	1.000	900.00	
	27		50.00	0.50	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	28		50.00	0.50	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	29		50.00	0.50	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00	1.000	900.00	
	30		50.00	0.50	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	31		50.00	0.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	32		50.00	0.00	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			

Example 2: Input file (cont.)

5	Element	3:	i	W(i)	t(i)	E(i)	G(i)	$\mu_1(i)$	$\mu_2(i)$	Ct(i)	Ftu(i)	[Eff	Fbru(i)]
	1		50.00	0.0	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	2		50.00	0.0	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	3		50.00	0.0	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	4		50.00	0.0	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00	0.000	900.00	
	5		50.00	0.0	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	6		50.00	0.70	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	7		50.00	0.70	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00	1.000	900.00	
	8		50.00	0.70	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	9		50.00	0.70	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	10		50.00	0.70	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00	1.000	900.00	
	11		50.00	0.70	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	12		50.00	0.70	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	13		50.00	0.70	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	14		50.00	0.70	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	15		50.00	0.70	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00	0.000	900.00	
	16		50.00	0.70	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00	0.000	900.00	
	17		50.00	0.70	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00	0.000	900.00	
	18		50.00	0.70	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00	0.000	900.00	
	19		50.00	0.70	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	20		50.00	0.70	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	21		50.00	0.70	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	22		50.00	0.70	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	23		50.00	0.70	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00	1.000	900.00	
	24		50.00	0.70	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	25		50.00	0.70	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	26		50.00	0.70	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00	1.000	900.00	
	27		50.00	0.70	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	28		50.00	0.0	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	29		50.00	0.0	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00	0.000	900.00	
	30		50.00	0.0	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	31		50.00	0.0	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			
	32		50.00	0.0	72400.0	27600.0	0.33	0.33	2.2500E-05	450.00			

Example 2: Input file (cont.)

```

5 Element 4: i W(i) t(i) E(i) G(i) p1(i) p2(i) Ct(i) Ftu(i) [Eff Fbru(i)]
  1 50.00 0.0 72400.0 27600.0 0.33 0.33 2.2500E-05 450.00
  2 50.00 0.0 72400.0 27600.0 0.33 0.33 2.2500E-05 450.00
  3 50.00 0.0 72400.0 27600.0 0.33 0.33 2.2500E-05 450.00
  4 50.00 0.0 72400.0 27600.0 0.33 0.33 2.2500E-05 450.00 0.000 900.00
  5 50.00 0.0 72400.0 27600.0 0.33 0.33 2.2500E-05 450.00
  6 50.00 0.0 72400.0 27600.0 0.33 0.33 2.2500E-05 450.00
  7 50.00 0.0 72400.0 27600.0 0.33 0.33 2.2500E-05 450.00 0.000 900.00
  8 50.00 0.0 72400.0 27600.0 0.33 0.33 2.2500E-05 450.00
  9 50.00 0.80 72400.0 27600.0 0.33 0.33 2.2500E-05 450.00
 10 50.00 0.80 72400.0 27600.0 0.33 0.33 2.2500E-05 450.00 1.000 900.00
 11 50.00 0.80 72400.0 27600.0 0.33 0.33 2.2500E-05 450.00
 12 50.00 0.80 72400.0 27600.0 0.33 0.33 2.2500E-05 450.00
 13 50.00 0.80 72400.0 27600.0 0.33 0.33 2.2500E-05 450.00
 14 50.00 0.80 72400.0 27600.0 0.33 0.33 2.2500E-05 450.00
 15 50.00 0.80 72400.0 27600.0 0.33 0.33 2.2500E-05 450.00 0.000 900.00
 16 50.00 0.80 72400.0 27600.0 0.33 0.33 2.2500E-05 450.00 0.000 900.00
 17 50.00 0.80 72400.0 27600.0 0.33 0.33 2.2500E-05 450.00 0.000 900.00
 18 50.00 0.80 72400.0 27600.0 0.33 0.33 2.2500E-05 450.00 0.000 900.00
 19 50.00 0.80 72400.0 27600.0 0.33 0.33 2.2500E-05 450.00
 20 50.00 0.80 72400.0 27600.0 0.33 0.33 2.2500E-05 450.00
 21 50.00 0.80 72400.0 27600.0 0.33 0.33 2.2500E-05 450.00
 22 50.00 0.80 72400.0 27600.0 0.33 0.33 2.2500E-05 450.00
 23 50.00 0.80 72400.0 27600.0 0.33 0.33 2.2500E-05 450.00 1.000 900.00
 24 50.00 0.80 72400.0 27600.0 0.33 0.33 2.2500E-05 450.00
 25 50.00 0.0 72400.0 27600.0 0.33 0.33 2.2500E-05 450.00
 26 50.00 0.0 72400.0 27600.0 0.33 0.33 2.2500E-05 450.00 0.000 900.00
 27 50.00 0.0 72400.0 27600.0 0.33 0.33 2.2500E-05 450.00
 28 50.00 0.0 72400.0 27600.0 0.33 0.33 2.2500E-05 450.00
 29 50.00 0.0 72400.0 27600.0 0.33 0.33 2.2500E-05 450.00 0.000 900.00
 30 50.00 0.0 72400.0 27600.0 0.33 0.33 2.2500E-05 450.00
 31 50.00 0.0 72400.0 27600.0 0.33 0.33 2.2500E-05 450.00
 32 50.00 0.0 72400.0 27600.0 0.33 0.33 2.2500E-05 450.00
6 Applied loads: Pleft(i) Pright(i) - Qleft(i) Qright(i) / DeltaT
 10000.0 10000.0 0.0 0.0
 0.0 0.0 0.0 0.0
 0.0 0.0 0.0 0.0
 0.0 0.0 0.0 0.0
 0.0

```


Example 2: Output file

Program MultiJoint
Validation example with four panels fastened between them. Upper one split in
Material: Aluminium 2024 T3.
Upper panel loaded at both ends.

Execution data:
=====

Date: 3-Jun-2020
MultiJoint 1.0. (c) Airbus Defence and Space. Jun-2020

Geometry:
=====

Point number	1	2	3	4	· · ·	Np-3	Np-2	Np-1	Np
Gr. of Fast. number:		1		2	· · ·	N-1		N	

Element 1: Pleft(1)	<----	#=====	#=====	#=====	#=	· · ·	#=====	#=====	#=====	#	---->	Pright(1)
Element 2: Pleft(2)	<----	#=====	#=====	#=====	#=	· · ·	#=====	#=====	#=====	#	---->	Pright(2)
· · · · ·												
· · · · ·												
Element n: Pleft(Np)	<----	#=====	#=====	#=====	#=	· · ·	#=====	#=====	#=====	#	---->	Pright(Np)

|-----> X

Applied loads (N):
=====

Axial:

Element 1	Pleft =	10000.0	Pright =	10000.0	Delta Temp =	0.00 °C
Element 2	Pleft =	0.0	Pright =	0.0		
Element 3	Pleft =	0.0	Pright =	0.0		
Element 4	Pleft =	0.0	Pright =	0.0		

Shear:

Element 1	Qleft =	0.0	Qright =	0.0
Element 2	Qleft =	0.0	Qright =	0.0
Element 3	Qleft =	0.0	Qright =	0.0
Element 4	Qleft =	0.0	Qright =	0.0

Example 2: Output file (cont.)

Materials properties and geometry:

=====

W <- Width
t <- Thickness
E <- Elastic modulus
G <- Shear modulus
 μ_{xy} <- Poisson ratio <xy>
 μ_{yx} <- Poisson ratio <yx>
Alfa <- Thermal expansion coefficient

Element 1 (Metallic)								
Pt	X (mm)	W (mm)	t (mm)	E (MPa)	G (MPa)	μ_{xy}	μ_{yx}	Alfa (mm/mm/C)
1	0.00	50.00	2.00	72400	27600	0.330	0.330	2.250E-05
2	10.00	50.00	2.00	72400	27600	0.330	0.330	2.250E-05
3	10.00	50.00	2.00	72400	27600	0.330	0.330	2.250E-05
4	20.00	50.00	2.00	72400	27600	0.330	0.330	2.250E-05
5	30.00	50.00	2.00	72400	27600	0.330	0.330	2.250E-05
6	30.00	50.00	2.00	72400	27600	0.330	0.330	2.250E-05
7	40.00	50.00	2.00	72400	27600	0.330	0.330	2.250E-05
8	50.00	50.00	2.00	72400	27600	0.330	0.330	2.250E-05
9	50.00	50.00	2.00	72400	27600	0.330	0.330	2.250E-05
10	60.00	50.00	2.00	72400	27600	0.330	0.330	2.250E-05
11	69.00	50.00	2.00	72400	27600	0.330	0.330	2.250E-05
12	69.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05
13	71.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05
14	71.00	50.00	2.00	72400	27600	0.330	0.330	2.250E-05
15	80.00	50.00	2.00	72400	27600	0.330	0.330	2.250E-05
16	100.00	50.00	2.00	72400	27600	0.330	0.330	2.250E-05
17	150.00	50.00	2.00	72400	27600	0.330	0.330	2.250E-05
18	170.00	50.00	2.00	72400	27600	0.330	0.330	2.250E-05
19	179.00	50.00	2.00	72400	27600	0.330	0.330	2.250E-05
20	179.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05
21	181.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05
22	181.00	50.00	2.00	72400	27600	0.330	0.330	2.250E-05
23	190.00	50.00	2.00	72400	27600	0.330	0.330	2.250E-05
24	200.00	50.00	2.00	72400	27600	0.330	0.330	2.250E-05
25	200.00	50.00	2.00	72400	27600	0.330	0.330	2.250E-05
26	210.00	50.00	2.00	72400	27600	0.330	0.330	2.250E-05
27	220.00	50.00	2.00	72400	27600	0.330	0.330	2.250E-05
28	220.00	50.00	2.00	72400	27600	0.330	0.330	2.250E-05
29	230.00	50.00	2.00	72400	27600	0.330	0.330	2.250E-05
30	240.00	50.00	2.00	72400	27600	0.330	0.330	2.250E-05
31	240.00	50.00	2.00	72400	27600	0.330	0.330	2.250E-05
32	250.00	50.00	2.00	72400	27600	0.330	0.330	2.250E-05

Example 2: Output file (cont.)

Pt	X (mm)	Element 2 (Metallic)						
		W (mm)	t (mm)	E (MPa)	G (MPa)	μ_{xy}	μ_{yx}	Alfa (mm/mm/C)
1	0.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05
2	10.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05
3	10.00	50.00	0.50	72400	27600	0.330	0.330	2.250E-05
4	20.00	50.00	0.50	72400	27600	0.330	0.330	2.250E-05
5	30.00	50.00	0.50	72400	27600	0.330	0.330	2.250E-05
6	30.00	50.00	0.50	72400	27600	0.330	0.330	2.250E-05
7	40.00	50.00	0.50	72400	27600	0.330	0.330	2.250E-05
8	50.00	50.00	0.50	72400	27600	0.330	0.330	2.250E-05
9	50.00	50.00	0.50	72400	27600	0.330	0.330	2.250E-05
10	60.00	50.00	0.50	72400	27600	0.330	0.330	2.250E-05
11	69.00	50.00	0.50	72400	27600	0.330	0.330	2.250E-05
12	69.00	50.00	0.50	72400	27600	0.330	0.330	2.250E-05
13	71.00	50.00	0.50	72400	27600	0.330	0.330	2.250E-05
14	71.00	50.00	0.50	72400	27600	0.330	0.330	2.250E-05
15	80.00	50.00	0.50	72400	27600	0.330	0.330	2.250E-05
16	100.00	50.00	0.50	72400	27600	0.330	0.330	2.250E-05
17	150.00	50.00	0.50	72400	27600	0.330	0.330	2.250E-05
18	170.00	50.00	0.50	72400	27600	0.330	0.330	2.250E-05
19	179.00	50.00	0.50	72400	27600	0.330	0.330	2.250E-05
20	179.00	50.00	0.50	72400	27600	0.330	0.330	2.250E-05
21	181.00	50.00	0.50	72400	27600	0.330	0.330	2.250E-05
22	181.00	50.00	0.50	72400	27600	0.330	0.330	2.250E-05
23	190.00	50.00	0.50	72400	27600	0.330	0.330	2.250E-05
24	200.00	50.00	0.50	72400	27600	0.330	0.330	2.250E-05
25	200.00	50.00	0.50	72400	27600	0.330	0.330	2.250E-05
26	210.00	50.00	0.50	72400	27600	0.330	0.330	2.250E-05
27	220.00	50.00	0.50	72400	27600	0.330	0.330	2.250E-05
28	220.00	50.00	0.50	72400	27600	0.330	0.330	2.250E-05
29	230.00	50.00	0.50	72400	27600	0.330	0.330	2.250E-05
30	240.00	50.00	0.50	72400	27600	0.330	0.330	2.250E-05
31	240.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05
32	250.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05

Example 2: Output file (cont.)

Element 3 (Metallic)								
Pt	X (mm)	W (mm)	t (mm)	E (MPa)	G (MPa)	μ_{xy}	μ_{yx}	Alfa (mm/mm/C)
1	0.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05
2	10.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05
3	10.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05
4	20.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05
5	30.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05
6	30.00	50.00	0.70	72400	27600	0.330	0.330	2.250E-05
7	40.00	50.00	0.70	72400	27600	0.330	0.330	2.250E-05
8	50.00	50.00	0.70	72400	27600	0.330	0.330	2.250E-05
9	50.00	50.00	0.70	72400	27600	0.330	0.330	2.250E-05
10	60.00	50.00	0.70	72400	27600	0.330	0.330	2.250E-05
11	69.00	50.00	0.70	72400	27600	0.330	0.330	2.250E-05
12	69.00	50.00	0.70	72400	27600	0.330	0.330	2.250E-05
13	71.00	50.00	0.70	72400	27600	0.330	0.330	2.250E-05
14	71.00	50.00	0.70	72400	27600	0.330	0.330	2.250E-05
15	80.00	50.00	0.70	72400	27600	0.330	0.330	2.250E-05
16	100.00	50.00	0.70	72400	27600	0.330	0.330	2.250E-05
17	150.00	50.00	0.70	72400	27600	0.330	0.330	2.250E-05
18	170.00	50.00	0.70	72400	27600	0.330	0.330	2.250E-05
19	179.00	50.00	0.70	72400	27600	0.330	0.330	2.250E-05
20	179.00	50.00	0.70	72400	27600	0.330	0.330	2.250E-05
21	181.00	50.00	0.70	72400	27600	0.330	0.330	2.250E-05
22	181.00	50.00	0.70	72400	27600	0.330	0.330	2.250E-05
23	190.00	50.00	0.70	72400	27600	0.330	0.330	2.250E-05
24	200.00	50.00	0.70	72400	27600	0.330	0.330	2.250E-05
25	200.00	50.00	0.70	72400	27600	0.330	0.330	2.250E-05
26	210.00	50.00	0.70	72400	27600	0.330	0.330	2.250E-05
27	220.00	50.00	0.70	72400	27600	0.330	0.330	2.250E-05
28	220.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05
29	230.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05
30	240.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05
31	240.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05
32	250.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05

Example 2: Output file (cont.)

Element 4 (Metallic)								
Pt	X (mm)	W (mm)	t (mm)	E (MPa)	G (MPa)	μ_{xy}	μ_{yx}	Alfa (mm/mm/C)
1	0.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05
2	10.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05
3	10.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05
4	20.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05
5	30.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05
6	30.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05
7	40.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05
8	50.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05
9	50.00	50.00	0.80	72400	27600	0.330	0.330	2.250E-05
10	60.00	50.00	0.80	72400	27600	0.330	0.330	2.250E-05
11	69.00	50.00	0.80	72400	27600	0.330	0.330	2.250E-05
12	69.00	50.00	0.80	72400	27600	0.330	0.330	2.250E-05
13	71.00	50.00	0.80	72400	27600	0.330	0.330	2.250E-05
14	71.00	50.00	0.80	72400	27600	0.330	0.330	2.250E-05
15	80.00	50.00	0.80	72400	27600	0.330	0.330	2.250E-05
16	100.00	50.00	0.80	72400	27600	0.330	0.330	2.250E-05
17	150.00	50.00	0.80	72400	27600	0.330	0.330	2.250E-05
18	170.00	50.00	0.80	72400	27600	0.330	0.330	2.250E-05
19	179.00	50.00	0.80	72400	27600	0.330	0.330	2.250E-05
20	179.00	50.00	0.80	72400	27600	0.330	0.330	2.250E-05
21	181.00	50.00	0.80	72400	27600	0.330	0.330	2.250E-05
22	181.00	50.00	0.80	72400	27600	0.330	0.330	2.250E-05
23	190.00	50.00	0.80	72400	27600	0.330	0.330	2.250E-05
24	200.00	50.00	0.80	72400	27600	0.330	0.330	2.250E-05
25	200.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05
26	210.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05
27	220.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05
28	220.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05
29	230.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05
30	240.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05
31	240.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05
32	250.00	50.00	0.00	72400	27600	0.330	0.330	2.250E-05

Example 2: Output file (cont.)

Allowable tensile and bearing values for elements:
=====

Element 1 (Metallic)					
Tensile			Bearing		
Pt	X (mm)	Max Str (MPa)	Gr	Nr	Fbru (MPa)
1	0.00	450.00	1	1	900.00
2	10.00	450.00	2	1	900.00
3	10.00	450.00	3	1	900.00
4	20.00	450.00	4	1	900.00
5	30.00	450.00	5	1	900.00
6	30.00	450.00	6	1	900.00
7	40.00	450.00	7	1	900.00
8	50.00	450.00	8	1	900.00
9	50.00	450.00	9	1	900.00
10	60.00	450.00	10	1	900.00
11	69.00	450.00			
12	69.00	450.00			
13	71.00	450.00			
14	71.00	450.00			
15	80.00	450.00			
16	100.00	450.00			
17	150.00	450.00			
18	170.00	450.00			
19	179.00	450.00			
20	179.00	450.00			
21	181.00	450.00			
22	181.00	450.00			
23	190.00	450.00			
24	200.00	450.00			
25	200.00	450.00			
26	210.00	450.00			
27	220.00	450.00			
28	220.00	450.00			
29	230.00	450.00			
30	240.00	450.00			
31	240.00	450.00			
32	250.00	450.00			

Example 2: Output file (cont.)

Element 2 (Metallic)					
Tensile			Bearing		
Pt	X (mm)	Max Str (MPa)	Gr	Nr	Fbru (MPa)
1	0.00	450.00	1	1	900.00
2	10.00	450.00	2	1	900.00
3	10.00	450.00	3	1	900.00
4	20.00	450.00	4	1	900.00
5	30.00	450.00	5	1	900.00
6	30.00	450.00	6	1	900.00
7	40.00	450.00	7	1	900.00
8	50.00	450.00	8	1	900.00
9	50.00	450.00	9	1	900.00
10	60.00	450.00	10	1	900.00
11	69.00	450.00			
12	69.00	450.00			
13	71.00	450.00			
14	71.00	450.00			
15	80.00	450.00			
16	100.00	450.00			
17	150.00	450.00			
18	170.00	450.00			
19	179.00	450.00			
20	179.00	450.00			
21	181.00	450.00			
22	181.00	450.00			
23	190.00	450.00			
24	200.00	450.00			
25	200.00	450.00			
26	210.00	450.00			
27	220.00	450.00			
28	220.00	450.00			
29	230.00	450.00			
30	240.00	450.00			
31	240.00	450.00			
32	250.00	450.00			

Example 2: Output file (cont.)

Element 3 (Metallic)					
Tensile			Bearing		
Pt	X (mm)	Max Str (MPa)	Gr	Nr	Fbru (MPa)
1	0.00	450.00	1	1	900.00
2	10.00	450.00	2	1	900.00
3	10.00	450.00	3	1	900.00
4	20.00	450.00	4	1	900.00
5	30.00	450.00	5	1	900.00
6	30.00	450.00	6	1	900.00
7	40.00	450.00	7	1	900.00
8	50.00	450.00	8	1	900.00
9	50.00	450.00	9	1	900.00
10	60.00	450.00	10	1	900.00
11	69.00	450.00			
12	69.00	450.00			
13	71.00	450.00			
14	71.00	450.00			
15	80.00	450.00			
16	100.00	450.00			
17	150.00	450.00			
18	170.00	450.00			
19	179.00	450.00			
20	179.00	450.00			
21	181.00	450.00			
22	181.00	450.00			
23	190.00	450.00			
24	200.00	450.00			
25	200.00	450.00			
26	210.00	450.00			
27	220.00	450.00			
28	220.00	450.00			
29	230.00	450.00			
30	240.00	450.00			
31	240.00	450.00			
32	250.00	450.00			

Example 2: Output file (cont.)

Element 4 (Metallic)					
Tensile			Bearing		
Pt	X (mm)	Max Str (MPa)	Gr	Nr	Fbru (MPa)
1	0.00	450.00	1	1	900.00
2	10.00	450.00	2	1	900.00
3	10.00	450.00	3	1	900.00
4	20.00	450.00	4	1	900.00
5	30.00	450.00	5	1	900.00
6	30.00	450.00	6	1	900.00
7	40.00	450.00	7	1	900.00
8	50.00	450.00	8	1	900.00
9	50.00	450.00	9	1	900.00
10	60.00	450.00	10	1	900.00
11	69.00	450.00			
12	69.00	450.00			
13	71.00	450.00			
14	71.00	450.00			
15	80.00	450.00			
16	100.00	450.00			
17	150.00	450.00			
18	170.00	450.00			
19	179.00	450.00			
20	179.00	450.00			
21	181.00	450.00			
22	181.00	450.00			
23	190.00	450.00			
24	200.00	450.00			
25	200.00	450.00			
26	210.00	450.00			
27	220.00	450.00			
28	220.00	450.00			
29	230.00	450.00			
30	240.00	450.00			
31	240.00	450.00			
32	250.00	450.00			

Example 2: Output file (cont.)

Fasteners locations: 10
=====

Gr	Pt	Nr	Delta X (mm)	Diam (mm)	Er (MPa)	Allowable (N)	Stiffness (N/mm)	Effectiveness (%)	Elem.
1	4	1	0.00	4.00	110000	33000.00	15771.80	100.0 - 100.0	1 - 2
							0.00	100.0 - 0.0	2 - 3
							0.00	0.0 - 0.0	3 - 4
2	7	1	0.00	4.00	110000	33000.00	15771.80	100.0 - 100.0	1 - 2
							15424.51	100.0 - 100.0	2 - 3
							0.00	100.0 - 0.0	3 - 4
3	10	1	0.00	4.00	110000	33000.00	15771.80	100.0 - 100.0	1 - 2
							15424.51	100.0 - 100.0	2 - 3
							18057.49	100.0 - 100.0	3 - 4
4	15	1	0.00	4.00	110000	33000.00	15771.80	100.0 - 100.0	1 - 2
							0.00	100.0 - 0.0	2 - 3
							0.00	0.0 - 0.0	3 - 4
5	16	1	0.00	4.00	110000	33000.00	15771.80	100.0 - 100.0	1 - 2
							0.00	100.0 - 0.0	2 - 3
							0.00	0.0 - 0.0	3 - 4
6	17	1	0.00	4.00	110000	33000.00	15771.80	100.0 - 100.0	1 - 2
							0.00	100.0 - 0.0	2 - 3
							0.00	0.0 - 0.0	3 - 4
7	18	1	0.00	4.00	110000	33000.00	15771.80	100.0 - 100.0	1 - 2
							0.00	100.0 - 0.0	2 - 3
							0.00	0.0 - 0.0	3 - 4
8	23	1	0.00	4.00	110000	33000.00	15771.80	100.0 - 100.0	1 - 2
							15424.51	100.0 - 100.0	2 - 3
							18057.49	100.0 - 100.0	3 - 4
9	26	1	0.00	4.00	110000	33000.00	15771.80	100.0 - 100.0	1 - 2
							15424.51	100.0 - 100.0	2 - 3
							0.00	100.0 - 0.0	3 - 4
10	29	1	0.00	4.00	110000	33000.00	15771.80	100.0 - 100.0	1 - 2
							0.00	100.0 - 0.0	2 - 3
							0.00	0.0 - 0.0	3 - 4

Example 2: Output file (cont.)

Groups of fasteners. Global data:
=====

Gr	Pt	C.G. (mm)	Stiffness (N/mm)	Elem.
---	---	---	---	---
1	4	20.00	15771.80	1 - 2
		20.00	0.00	2 - 3
		20.00	0.00	3 - 4
2	7	40.00	15771.80	1 - 2
		40.00	15424.51	2 - 3
		40.00	0.00	3 - 4
3	10	60.00	15771.80	1 - 2
		60.00	15424.51	2 - 3
		60.00	18057.49	3 - 4
4	15	80.00	15771.80	1 - 2
		80.00	0.00	2 - 3
		80.00	0.00	3 - 4
5	16	100.00	15771.80	1 - 2
		100.00	0.00	2 - 3
		100.00	0.00	3 - 4
6	17	150.00	15771.80	1 - 2
		150.00	0.00	2 - 3
		150.00	0.00	3 - 4
7	18	170.00	15771.80	1 - 2
		170.00	0.00	2 - 3
		170.00	0.00	3 - 4
8	23	190.00	15771.80	1 - 2
		190.00	15424.51	2 - 3
		190.00	18057.49	3 - 4
9	26	210.00	15771.80	1 - 2
		210.00	15424.51	2 - 3
		210.00	0.00	3 - 4
10	29	230.00	15771.80	1 - 2
		230.00	0.00	2 - 3
		230.00	0.00	3 - 4

Example 2: Output file (cont.)

Load transfer due to applied loads:

Results shown partially: Only for elements 1 and 2
Only for points 1 to 21

X (mm)	Element 1 (Metallic)			Fasteners			Element 2 (Metallic)		
	Load (N)	Axial-Str (MPa)	Shear-Str (MPa)	Bearing (Mpa)	Load (N)	Bearing (MPa)	Load (N)	Axial-Str (MPa)	Shear-Str (MPa)
0.00	10000.0	100.00	0.00				0.0	0.00	0.00
	10000.0	100.00	0.00				0.0	0.00	0.00
	10000.0	100.00	0.00				0.0	0.00	0.00
10.00	10000.0	100.00	0.00				0.0	0.00	0.00
	10000.0	100.00	0.00				0.0	0.00	0.00
10.00	10000.0	100.00	0.00				0.0	0.00	0.00
	10000.0	100.00	0.00				0.0	0.00	0.00
20.00	10000.0	100.00	0.00				0.0	0.00	0.00
	10000.0	100.00	0.00				0.0	0.00	0.00
20.00	6997.8	69.98	0.00	375.28	3002.2	1501.11	3002.2	120.09	0.00
	6997.8	69.98	0.00				3002.2	120.09	0.00
30.00	6997.8	69.98	0.00				3002.2	120.09	0.00
	6997.8	69.98	0.00				3002.2	120.09	0.00
30.00	6997.8	69.98	0.00				3002.2	120.09	0.00
	6997.8	69.98	0.00				3002.2	120.09	0.00
40.00	6997.8	69.98	0.00				3002.2	120.09	0.00
	6997.8	69.98	0.00				3002.2	120.09	0.00
40.00	3777.2	37.77	0.00	402.57	3220.6	1610.28	4138.5	165.54	0.00
	3777.2	37.77	0.00				4138.5	165.54	0.00
50.00	3777.2	37.77	0.00				4138.5	165.54	0.00
	3777.2	37.77	0.00				4138.5	165.54	0.00
50.00	3777.2	37.77	0.00				4138.5	165.54	0.00
	3777.2	37.77	0.00				4138.5	165.54	0.00
60.00	0.0	0.00	0.00	472.15	3777.2	1888.61	6282.9	251.32	0.00
	0.0	0.00	0.00				6282.9	251.32	0.00
69.00	0.0	0.00	0.00				6282.9	251.32	0.00
	0.0	0.00	0.00				6282.9	251.32	0.00
69.00	0.0	0.00	0.00				6282.9	251.32	0.00
	0.0	0.00	0.00				6282.9	251.32	0.00
71.00	0.0	0.00	0.00				6282.9	251.32	0.00
	0.0	0.00	0.00				6282.9	251.32	0.00
71.00	0.0	0.00	0.00				6282.9	251.32	0.00
	0.0	0.00	0.00				6282.9	251.32	0.00
80.00	1516.0	15.16	0.00	189.50	1516.0	758.00	4766.9	190.68	0.00
	1516.0	15.16	0.00				4766.9	190.68	0.00
100.00	2267.3	22.67	0.00	93.91	751.3	375.65	4015.6	160.63	0.00
	2267.3	22.67	0.00				4015.6	160.63	0.00
150.00	1516.0	15.16	0.00	93.91	751.3	375.65	4766.9	190.68	0.00
	1516.0	15.16	0.00				4766.9	190.68	0.00
170.00	0.0	0.00	0.00	189.50	1516.0	758.00	6282.9	251.32	0.00
	0.0	0.00	0.00				6282.9	251.32	0.00
179.00	0.0	0.00	0.00				6282.9	251.32	0.00
	0.0	0.00	0.00				6282.9	251.32	0.00
179.00	0.0	0.00	0.00				6282.9	251.32	0.00
	0.0	0.00	0.00				6282.9	251.32	0.00
181.00	0.0	0.00	0.00				6282.9	251.32	0.00
	0.0	0.00	0.00				6282.9	251.32	0.00

Example 2: Output file (cont.)

Reserve factors for applied loads:
=====

Unfastened areas:

	Element 1	Element 2	Element 3	Element 4
	-----	-----	-----	-----
Material:	Metallic	Metallic	Metallic	Metallic
Most loaded point:	250.00 mm	190.00 mm	190.00 mm	190.00 mm
Von-Mises stress:	100.00 MPa	251.32 MPa	79.87 MPa	23.04 MPa
Maximum strain:				
Allowable Stress/strain:	450.00 MPa	450.00 MPa	450.00 MPa	450.00 MPa
Reserve factor:	>2.00	1.79	>2.00	>2.00

Fastened areas:

		Element 1		
		-----	-----	-----
Gr	Nr	Bearing (MPa)	Fbru (MPa)	R.F.
---	---	-----	-----	-----
1	1	375.28	900.00	>2.00
2	1	402.57	900.00	>2.00
3	1	472.15	900.00	1.90
4	1	189.50	900.00	>2.00
5	1	93.91	900.00	>2.00
6	1	93.91	900.00	>2.00
7	1	189.50	900.00	>2.00
8	1	472.15	900.00	1.90
9	1	402.57	900.00	>2.00
10	1	375.28	900.00	>2.00

		Element 2		
		-----	-----	-----
Gr	Nr	Bearing (MPa)	Fbru (MPa)	R.F.
---	---	-----	-----	-----
1	1	1501.11	900.00	0.59
2	1	1918.10	900.00	0.46
3	1	2057.50	900.00	0.43
4	1	758.00	900.00	1.18
5	1	375.65	900.00	>2.00
6	1	375.65	900.00	>2.00
7	1	758.00	900.00	1.18
8	1	2057.50	900.00	0.43
9	1	1918.10	900.00	0.46
10	1	1501.11	900.00	0.59

Example 2: Output file (cont.)

Element 3				
Gr	Nr	Bearing (MPa)	Fbru (MPa)	R.F.
1	1	0.00	900.00	>2.00
2	1	744.40	900.00	1.20
3	1	669.57	900.00	1.34
4	1	0.00	900.00	>2.00
5	1	0.00	900.00	>2.00
6	1	0.00	900.00	>2.00
7	1	0.00	900.00	>2.00
8	1	669.57	900.00	1.34
9	1	744.40	900.00	1.20
10	1	0.00	900.00	>2.00

Element 4				
Gr	Nr	Bearing (MPa)	Fbru (MPa)	R.F.
1	1	0.00	900.00	>2.00
2	1	0.00	900.00	>2.00
3	1	287.95	900.00	>2.00
4	1	0.00	900.00	>2.00
5	1	0.00	900.00	>2.00
6	1	0.00	900.00	>2.00
7	1	0.00	900.00	>2.00
8	1	287.95	900.00	>2.00
9	1	0.00	900.00	>2.00
10	1	0.00	900.00	>2.00

Example 2: Output file (cont.)

Reserve factors for fasteners:
=====

Gr	Nr	Joint	Shear (N)	Allowable (N)	R.F.
---	---	---	-----	-----	-----
1	1	1-2	3002.2	33000.0	>2.00
		2-3	0.0	33000.0	>2.00
		3-4	0.0	33000.0	>2.00
2	1	1-2	3220.6	33000.0	>2.00
		2-3	2084.3	33000.0	>2.00
		3-4	0.0	33000.0	>2.00
3	1	1-2	3777.2	33000.0	>2.00
		2-3	1632.7	33000.0	>2.00
		3-4	921.4	33000.0	>2.00
4	1	1-2	1516.0	33000.0	>2.00
		2-3	0.0	33000.0	>2.00
		3-4	0.0	33000.0	>2.00
5	1	1-2	751.3	33000.0	>2.00
		2-3	0.0	33000.0	>2.00
		3-4	0.0	33000.0	>2.00
6	1	1-2	751.3	33000.0	>2.00
		2-3	0.0	33000.0	>2.00
		3-4	0.0	33000.0	>2.00
7	1	1-2	1516.0	33000.0	>2.00
		2-3	0.0	33000.0	>2.00
		3-4	0.0	33000.0	>2.00
8	1	1-2	3777.2	33000.0	>2.00
		2-3	1632.7	33000.0	>2.00
		3-4	921.4	33000.0	>2.00
9	1	1-2	3220.6	33000.0	>2.00
		2-3	2084.3	33000.0	>2.00
		3-4	0.0	33000.0	>2.00
10	1	1-2	3002.2	33000.0	>2.00
		2-3	0.0	33000.0	>2.00
		3-4	0.0	33000.0	>2.00

7 Program validation

In order to validate the program, the example 2 have been modelled and solved with MSC/NASTRAN code, version 2019.0.0, using the solution 101 (linear).

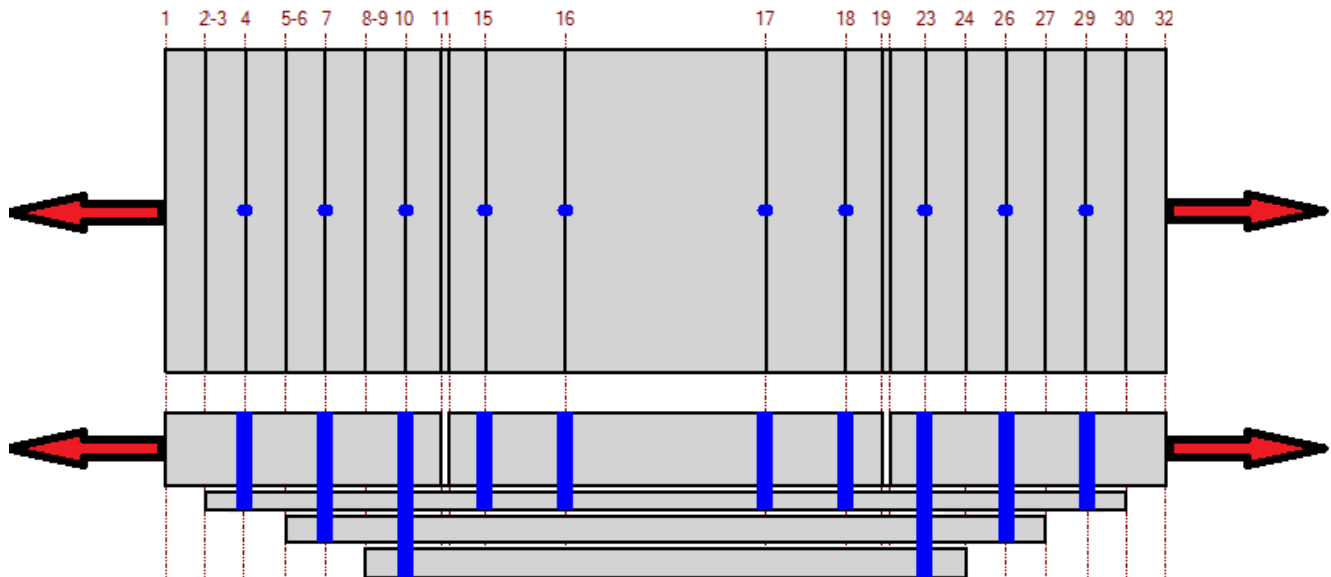


Figure 18: Example for program validation

Each one of the four elements between two non-coincident points have been modelled by means of ROD elements working in traction/compression, with a constant area ($W \cdot t$) each one.

Of course, the upper panel represents three elements with no continuity between them.

The fasteners have been modelled by means of ELAS1 elements, with the same stiffness between each two consecutive panels as calculated by MULTIJOINT.

The total number of elements in the Nastran bulk data file is:

Type of element	Quantity
GRID	70
CROD	64
PROD	64
MAT1	1
CELAS1	16
PELAS	16
FORCE	1
PARAM	2
SPC1	1

Comparing outputs results from MULTIJOINT and NASTRAN in terms of internal loads in elements as well as transferred loads through the fasteners, the relative differences are less than 0.006% in all cases.

The results from both codes, MULTIJOINT and MSC/NASTRAN are shown on tables below:

Transferred loads by fasteners:

Fastener	Elements	MULTIJOINT	NASTRAN	Diff. (%)
1	1-2	3002,2	-3002,23	0,0009%
	2-3	0,0	---	
	3-4	0,0	---	
2	1-2	3220,6	-3220,56	-0,0014%
	2-3	2084,3	-2084,32	0,0012%
	3-4	0,0	---	
3	1-2	3777,2	-3777,22	0,0004%
	2-3	1632,7	-1632,73	0,0016%
	3-4	921,4	-921,45	0,0054%
4	1-2	1516,0	1516,01	0,0003%
	2-3	0,0	---	
	3-4	0,0	---	
5	1-2	751,3	751,30	0,0000%
	2-3	0,0	---	
	3-4	0,0	---	
6	1-2	751,3	-751,30	0,0000%
	2-3	0,0	---	
	3-4	0,0	---	
7	1-2	1516,0	-1516,01	0,0003%
	2-3	0,0	---	
	3-4	0,0	---	
8	1-2	3777,2	3777,22	0,0004%
	2-3	1632,7	1632,73	0,0016%
	3-4	921,4	921,45	0,0054%
9	1-2	3220,6	3220,56	-0,0014%
	2-3	2084,3	2084,32	0,0012%
	3-4	0,0	---	
10	1-2	3002,2	3002,23	0,0009%
	2-3	0,0	---	
	3-4	0,0	---	

Loads in elements:

Segment	Element 1			Element 2		
	MULTIJOINT	NASTRAN	Diff. (%)	MULTIJOINT	NASTRAN	Diff. (%)
1-2	10000,0	10000,00	0,0000%	0,0	---	
2-3	10000,0	---		0,0	---	
3-4	10000,0	10000,00	0,0000%	0,0	0,00	
4-5	6997,8	6997,77	-0,0004%	3002,2	3002,23	0,0009%
5-6	6997,8	---		3002,2	---	
6-7	6997,8	6997,77	-0,0004%	3002,2	3002,23	0,0009%
7-8	3777,2	3777,22	0,0004%	4138,5	4138,46	-0,0009%
8-9	3777,2	---		4138,5	---	
9-10	3777,2	3777,22	0,0004%	4138,5	4138,46	-0,0009%
10-11	0,0	0,00		6282,9	6282,95	0,0008%
11-12	0,0	---		6282,9	---	
12-13	0,0	---		6282,9	6282,95	0,0008%
13-14	0,0	---		6282,9	---	
14-15	0,0	0,00		6282,9	6282,95	0,0008%
15-16	1516,0	1516,01	0,0003%	4766,9	4766,95	0,0010%
16-17	2267,3	2267,30	0,0002%	4015,6	4015,65	0,0011%
17-18	1516,0	1516,01	0,0003%	4766,9	4766,95	0,0010%
18-19	0,0	0,00		6282,9	6282,95	0,0008%
19-20	0,0	---		6282,9	---	
20-21	0,0	---		6282,9	6282,95	0,0008%
21-22	0,0	---		6282,9	---	
22-23	0,0	0,00		6282,9	6282,95	0,0008%
23-24	3777,2	3777,22	0,0004%	4138,5	4138,46	-0,0009%
24-25	3777,2	---		4138,5	---	
25-26	3777,2	3777,22	0,0004%	4138,5	4138,46	-0,0009%
26-27	6997,8	6997,77	-0,0004%	3002,2	3002,23	0,0009%
27-28	6997,8	---		3002,2	---	
28-29	6997,8	6997,77	-0,0004%	3002,2	3002,23	0,0009%
29-30	10000,0	10000,00	0,0000%	0,0	0,00	
30-31	10000,0	---		0,0	---	
31-32	10000,0	10000,00	0,00	0,0	---	

Loads in elements: (Cont.)

Segment	Element 3			Element 4		
	MULTIJOINT	NASTRAN	Diff. (%)	MULTIJOINT	NASTRAN	Diff. (%)
1-2	0,0	---		0,0	---	
2-3	0,0	---		0,0	---	
3-4	0,0	---		0,0	---	
4-5	0,0	---		0,0	---	
5-6	0,0	---		0,0	---	
6-7	0,0	0,00		0,0	---	
7-8	2084,3	2084,32	0,0012%	0,0	---	
8-9	2084,3	---		0,0	---	
9-10	2084,3	2084,32	0,0012%	0,0	0,00	
10-11	2795,6	2795,60	0,0000%	921,4	921,45	0,0054%
11-12	2795,6	---		921,4	---	
12-13	2795,6	2795,60	0,0000%	921,4	921,45	0,0054%
13-14	2795,6	---		921,4	---	
14-15	2795,6	2795,60	0,0000%	921,4	921,45	0,0054%
15-16	2795,6	2795,60	0,0000%	921,4	921,45	0,0054%
16-17	2795,6	2795,60	0,0000%	921,4	921,45	0,0054%
17-18	2795,6	2795,60	0,0000%	921,4	921,45	0,0054%
18-19	2795,6	2795,60	0,0000%	921,4	921,45	0,0054%
19-20	2795,6	---		921,4	---	
20-21	2795,6	2795,60	0,0000%	921,4	921,45	0,0054%
21-22	2795,6	---		921,4	---	
22-23	2795,6	2795,60	0,0000%	921,4	921,45	0,0054%
23-24	2084,3	2084,32	0,0012%	0,0	0,00	
24-25	2084,3	---		0,0	---	
25-26	2084,3	2084,32	0,0012%	0,0	---	
26-27	0,0	0,00		0,0	---	
27-28	0,0	---		0,0	---	
28-29	0,0	---		0,0	---	
29-30	0,0	---		0,0	---	
30-31	0,0	---		0,0	---	
31-32	0,0	---		0,0	---	